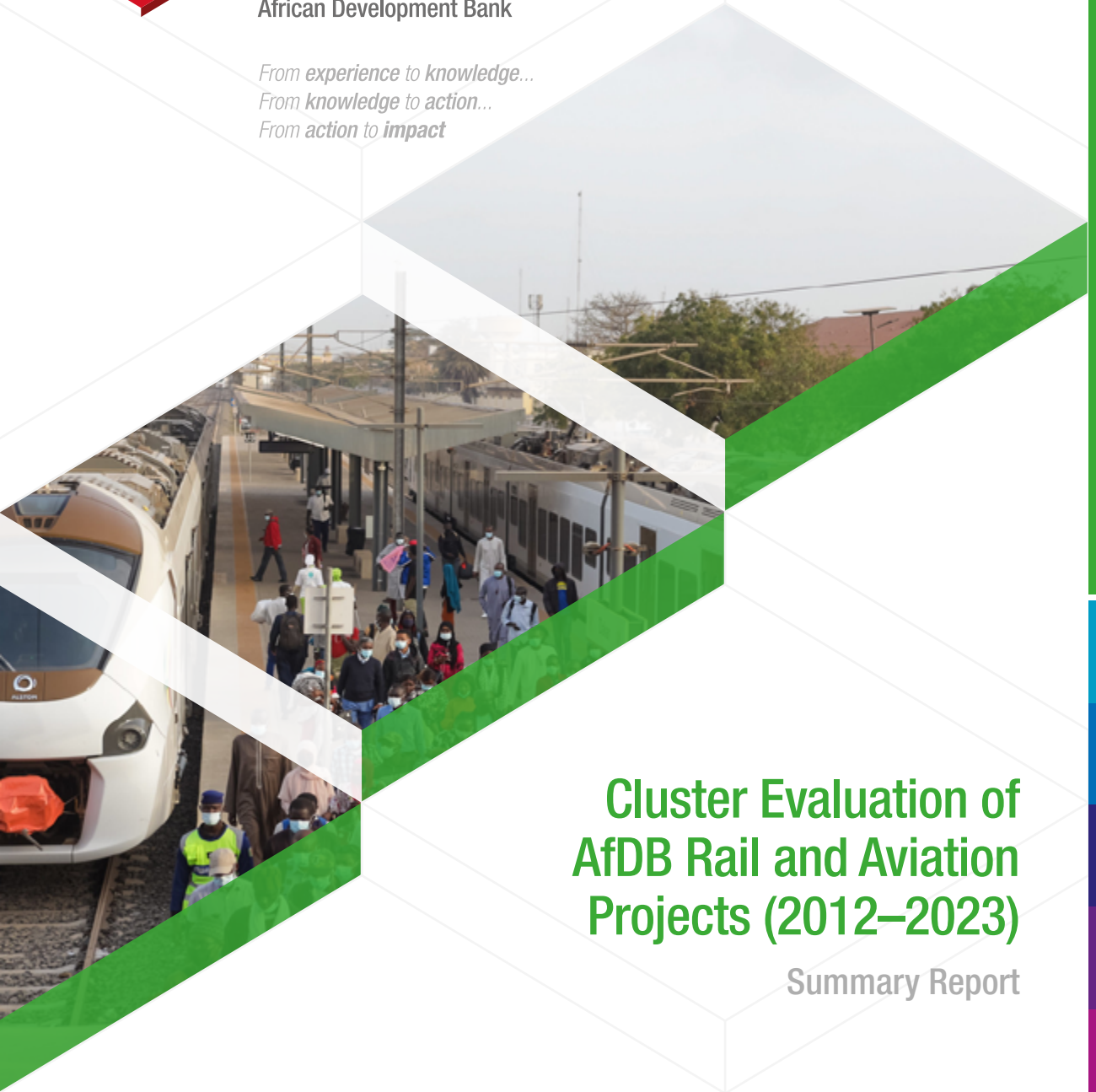


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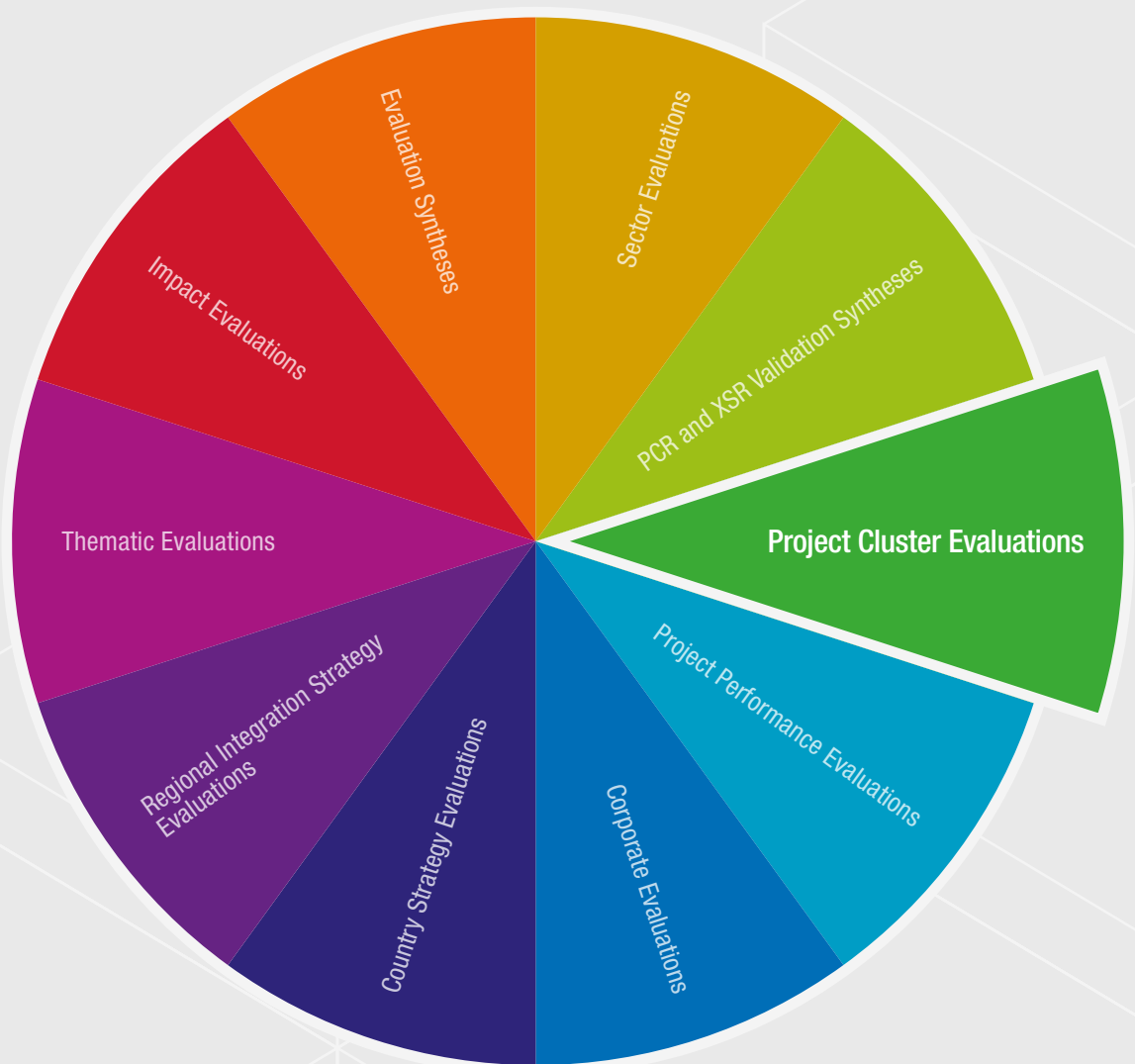
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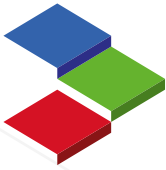


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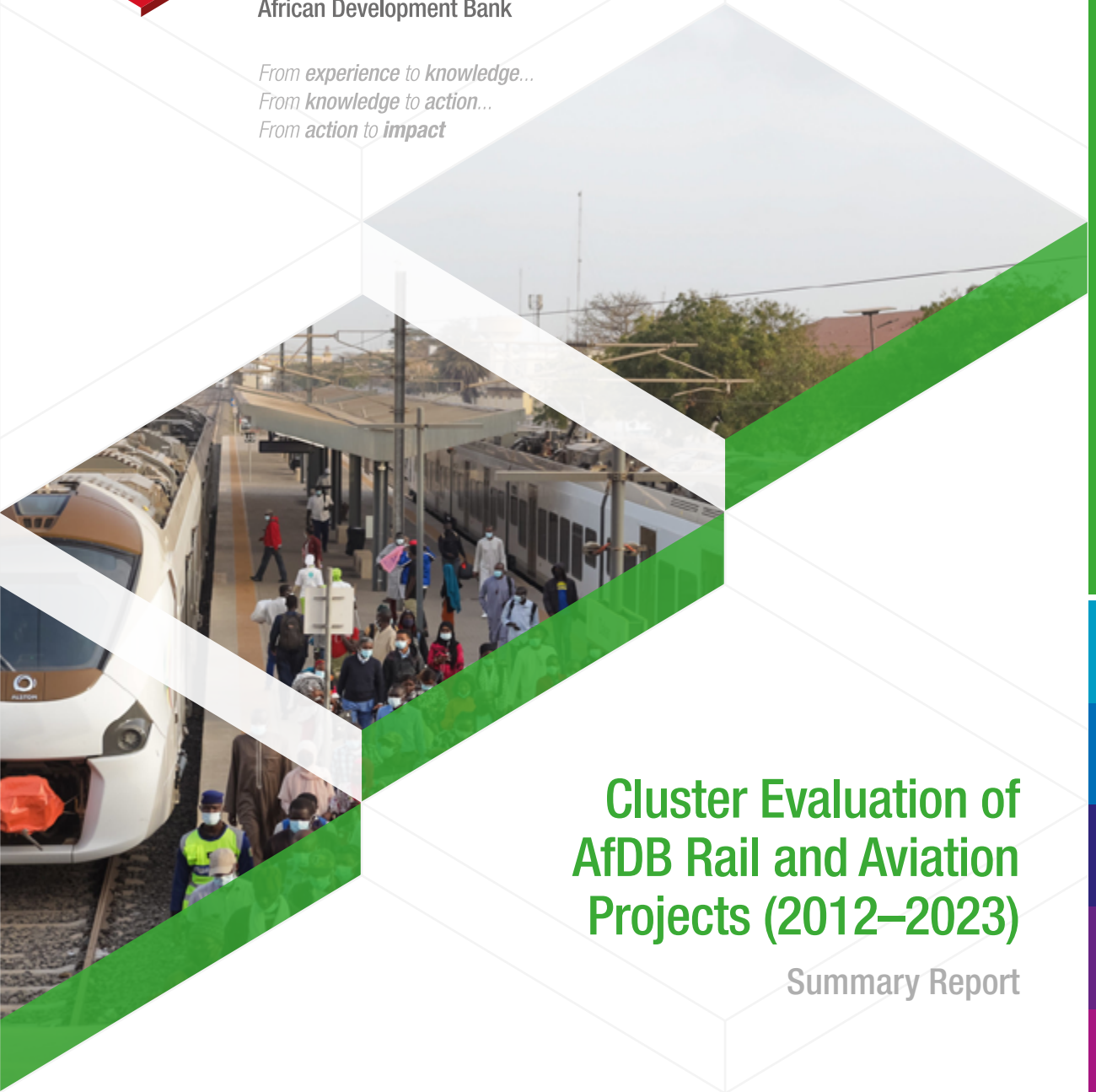




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Cluster Evaluation of AfDB Rail and Aviation Projects (2012–2023)

Summary Report



AFRICAN DEVELOPMENT BANK GROUP

May 2025

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The overarching objective of the African Development Bank Group is to spur sustainable economic development and social progress in its regional member countries (RMCs), thus contributing to poverty reduction. The Bank Group achieves this objective by mobilizing and allocating resources for investment in RMCs and providing policy advice and technical assistance to support development efforts.

About Independent Development Evaluation (IDEV)

The mission of Independent Development Evaluation at the AfDB is to enhance the development effectiveness of the institution in its regional member countries through independent and instrumental evaluations and partnerships for sharing knowledge.

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Abbreviations and Acronyms

AFCAC	African Civil Aviation Commission		
AfDB	African Development Bank Group	IPR	Implementation Progress and Results Report
AIBD	Blaise Diagne International Airport	M&E	Monitoring and Evaluation
ARMFA	Africa Road Maintenance Funds Association	OEM	Original Equipment Manufacturer
AU	African Union	PAR	Project Appraisal Report
AUC	African Union Commission	PASTA-CO	Air Transport Sectors of West and Central Africa Project
BTOR	Back-to-Office Report	PCR	Project Completion Report
COMESA	Common Market for Eastern and Southern Africa	PL	Project Loan
CSP	Country Strategy Paper	RISP	Regional Integration Strategy Paper
ECA	Economic Commission for Africa	RMC	Regional Member Country
ECOWAS	Economic Community of West African States	SAATM	Single African Air Transport Market
EIRR	Economic Internal Rate of Return	SDGs	Sustainable Development Goals
ENPV	Economic Net Present Value	SGR	Standard Gauge Railway
EQ	Evaluation Question	ToC	Theory of Change
EU	European Union	TYS	Ten-Year Strategy
GACL	Ghana Airports Company Limited	UA*	Unit of Account
GHG	Greenhouse Gas	UK	United Kingdom
HIV/AIDS	Human Immunodeficiency Virus / Acquired Immune Deficiency Syndrome	USD	United States Dollar
ICAO	International Civil Aviation Organization	WAEMU	West African Economic and Monetary Union
IDEV	Independent Development Evaluation (at the AfDB)	XSR	Expanded Supervision Report

* 1 UA = 1.31391 USD as of 30th November 2024.



Executive Summary

This summary report presents the key findings of a cluster evaluation conducted by Independent Development Evaluation (IDEV) of 16 rail and air transport projects supported by the African Development Bank Group (AfDB or “the Bank”) across the African continent between 2012 and 2023.

Owing to the unmet demand for adequate transport services on the continent and the sector’s ability to foster inclusive economic transformation, facilitate local and regional trade, and alleviate poverty, the transport sector is a strategic focal area for the Bank and its Regional Member Countries (RMCs). This is evidenced by the commitment of a significant share of the Bank’s resources to investments in transport infrastructure projects as well as institutional support and technical assistance across the continent. Within the transport sector, the Bank distinguishes four sub-sectors: road, aviation, rail, and maritime transport. IDEV previously conducted a [cluster evaluation of AfDB road and port projects](#) (in 2021). The current cluster evaluation of rail and aviation projects complements the cluster evaluation of road and port projects, and together they provide specific inputs to IDEV’s broader [evaluation of the Bank’s support for the transport sector](#) as a whole (2012-2023).

Between 2012 and 2023, the AfDB financed 11 railway and 25 aviation projects with total net funding of Units of Account (UA) 389.11 million (USD 518.15 million) and over UA 756.14 million (USD 1 billion) respectively. While the aviation sub-sector saw more projects, the average funding per project was higher for railways UA 35.37 million (USD 47.1 million) compared to aviation UA 30.25 million (USD 40.3 million). The rail and aviation sub-sectors accounted for only 2.8% and 5.4% respectively of the Bank’s total transport sector funding over the period, highlighting their relatively low prioritization compared to other sub-sectors like roads and maritime transport.

This evaluation focused on 16 of the 36 approved rail and aviation projects, and assessed them in terms of their relevance, coherence, efficiency, effectiveness, and sustainability. It also assessed the additionality of five of the projects, which were non-sovereign operations. It made a number of observations related to the planning, design, management and implementation of rail and aviation projects in the context of Africa and identified several pertinent lessons to inform the way forward for the Bank’s support for these two sub-sectors.

Methodology

Using the framework established by a transport sector Theory of Change and applying the international evaluation criteria of relevance, coherence, effectiveness, efficiency, and sustainability, the evaluation followed a mixed-methods approach and brought together findings from (i) a portfolio review of the Bank’s investments in the transport sector made between 2012 and 2023; (ii) in-depth case studies of eight of the Bank’s rail and aviation projects approved and/or completed between 2012 and 2023; and (iii) a desk-based review of a further eight of the Bank’s rail and aviation projects approved and/or completed between 2012 and 2023.

Data sources underpinning these findings included: (i) information from the Bank’s internal databases; (ii) documentary sources, including project appraisal reports, project completion reports, project completion report evaluation notes, implementation progress and results reports (IPRs), back-to-office reports, independent studies, country strategy papers (CSPs), regional integration strategy papers (RISPs), sector strategies and other relevant publicly available sources; (iii) interviews with project stakeholders, including executing agencies, Bank task managers, project beneficiaries, development

partners, government ministries and other public agencies; and (iv) site visits to the main physical infrastructure and ancillary project components for the eight in-depth case studies. The evaluation applied a four-point rating scale¹ to attribute a rating for each evaluation criterion (see Annex 2).

Main Findings

Relevance: To what extent did the objectives and design of the Bank's interventions respond to beneficiaries' global, country, and partner/institution needs?

The evaluation found the objectives of the cluster projects to be well aligned with the relevant strategies and priorities of the Bank and other relevant entities. This included the Bank's Ten-Year Strategy (TYS), the High 5s, CSPs, and RISPs, as well as the United Nations Sustainable Development Goals and the African Union's Agenda 2063. The favourable performance in this respect suggests that the Bank's selection criteria ensure that the transport projects it funds are - at the conceptual level- responsive to the specific needs of RMCs and contribute to the broader vision that the Bank has for socio-economic development on the continent. For example, the objectives of the multinational institutional support to the African Civil Aviation Commission for the implementation of the Single African Air Transport Market were well-aligned with the AfDB's TYS (2013-2022) to drive inclusive and green growth on the African continent, as well as its High 5s - particularly *Integrate Africa* and *Improving the quality of life of people in Africa*.

The evaluation found that **despite most cluster projects having cogent logical frameworks, some challenges related to project design were observed, with consequences for the relative success and timeliness of project delivery.** These included shortcomings related to the clarity of project objectives, risk identification and mitigation, integration of lessons learned from previous projects (whether Bank-financed or otherwise) and

monitoring of implementation progress. Although projects' ancillary components were typically found to have clear complementarity with their associated core components, their planning was generally less comprehensive, contributing to delivery shortcomings. Examples include the feasibility studies on the extension of the Cameroon-Chad Railway and on the Ethiopia-Sudan Standard Gauge Railway.

Overall, the **relevance** of the cluster projects was assessed to be **satisfactory**.

Coherence: To what extent did other Bank interventions support or undermine the Bank's interventions in the rail and air transport sub-sectors and vice versa? And to what extent were the AfDB's interventions complementary, harmonized and coordinated with other development partners' support?

The evaluation found the internal coherence within both the rail and aviation sub-sectors to be strong. All the cluster projects were found to have robust synergies and interlinkages with other projects in the indicative operational program of the CSPs and/or the existing portfolio in the case study countries. For example, the cluster projects with complementary initiatives supported the agriculture sector and facilitated access to markets, such as the Nacala Rail Corridor Project and Port Value Addition Project, which allowed the Bank to leverage its investments in physical infrastructure to contribute to the Feed Africa priority. Also, the Bank's multinational investments in the two sub-sectors consisted of trans-border international airports and multinational rail projects, which supported the regional integration agenda. In particular, the Bank's interventions in Southern Africa between 2012 and 2023 were found broadly consistent with the goals set out in the two RISPs covering the period — regional infrastructure forms a pillar of the strategy in each of these RISPs, with an emphasis on transport.

The evaluation also found evidence of external coherence with the interventions of other

development partners. For example, the Project for Support to the Air Transport Sectors of West and Central Africa (PASTA-CO) was found to have strong coherence with the European Union-Africa Safety in Aviation project to build the capacity of Regional Safety Oversight Organizations in Africa.

Both **internal and external coherence** were found **satisfactory** for all the projects in the cluster.

Effectiveness: How effective was the Bank in achieving its policy and strategic objectives and results (outputs and outcomes) regarding the two sub-sectors?

While performance in terms of relevance and coherence, which principally relate to projects' pre-implementation stages, was generally deemed satisfactory, the cluster projects were found to have mixed levels of success during implementation, as reflected in their attainment of planned outputs. Although often delayed, projects that centred around the development of physical transport infrastructure generally delivered their core components to appropriate design specifications and standards, with operating performance achieving planned levels of service. Examples of this included the Ghana Airports Company Limited (GACL) Capital Investment Programme, and the Railway Infrastructure Reinforcement project in Morocco. In contrast, projects' ancillary components were only partly delivered, often towards the end of implementation schedules, or cancelled altogether. For instance, the non-implementation of the activities under the "consolidation of gender achievements" component of the Railway Infrastructure Reinforcement Project in Morocco was explained by the Moroccan National Railways Office's inability to conclude an agreement with local authorities towards the implementation of the planned activities. Similarly, under the Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I, the intended socio-economic infrastructure and facilities had not been completed and were at risk of being cancelled at completion. Projects that focused on institutional capacity building were found

to have limited success in achieving their intended outputs, with capacity limitations of the executing agencies generally being a core constraint.

In terms of outcomes, the evaluation found that the development of physical infrastructure schemes typically had direct benefits for users in terms of connectivity and operational efficiency and, where relevant, safety improvements. For example, although the Dakar-Diamniadio-AIBD Regional Express Train Project – Phase I covered a relatively localized area of influence, its connection to Senegal's Blaise Diagne International Airport (AIBD) in the second phase of the project had a direct impact on Dakar's connectivity. Despite the successful attainment of outcomes, some projects highlighted limitations relating to the monitoring and recording of objectives, as well as the objectives themselves. The cluster projects centred around institutional capacity building were found to be less successful in terms of their outcome measures.

Overall, the evaluation rated the **effectiveness** of the cluster projects as **satisfactory**.

Efficiency: How efficient (timeliness and resource use) was the Bank in delivering its support to the two sub-sectors in RMCs?

In terms of time efficiency, a review of the IPRs of the cluster projects noted long delays experienced during the delivery of most projects, with a number of sources identified. For instance, 14 of the 16 cluster projects recorded delays of between 17 and 60 months beyond their scheduled implementation periods. The reasons identified for the delays included capacity limitations of executing agencies; limited flexibility in implementing Bank processes, with requirements that were challenging to adhere to — most notably the need for non-objection notices for all project changes; and ineffective coordination with delivery partners and stakeholders. The delays were not limited to the rail and aviation cluster projects, with the portfolio analysis showcasing a high frequency of delays across the Bank's wider transport portfolio.

It was also noted from the review of IPRs that despite the implementation delays, most projects were viewed favourably in terms of their use of financial resources, as in the case of the Corporate Loan to Ethiopian Airlines Project. However, some significant cost overruns, financial management challenges, and breaches of Bank regulations were also noted in the project assessments. These shortcomings were typically traced back to changes in project designs, limited guidance and supervision of executing agencies, and their ability to comply with the Bank's procedural requirements, whose implementation was considered by beneficiaries as less flexible and not commensurate with the scale, complexity and significance of projects.

Overall, the evaluation assessed the **efficiency** of the cluster projects as **partly unsatisfactory**.

Sustainability: How sustainable were the development results?

The ability of the interventions to provide long-term benefits was found mixed, with a large degree of variance noted between assessments of sustainability. From a technical perspective, most physical infrastructure designs appeared to reflect appropriate quality standards and adaptation elements. However, long-term **financial sustainability** concerns were raised across several project assessments, with potential consequences for infrastructure maintenance. These concerns stemmed from the projects' limited revenue-generating potential and the associated reliance on subsidization for cost recovery, as well as deficient financial management practices by some executing agencies during project implementation. Relatedly, significant differences in the **institutional capacities** and stability of executing agencies and other relevant participants were observed, suggesting the need for the Bank to better gauge the degree of supervision and implementation support needed on a project-by-project basis. The evaluation highlighted significant **ownership** by the beneficiary institutions and governments, noting cases of

clear mandates and effective policies for rail and aviation maintenance, planning, and organization, and beneficiaries showing a strong commitment to sharing project costs. However, continuity risks were highlighted for ancillary components aimed at advancing **environmental and social agendas**. Project assessments found little evidence of arrangements in place to ensure adequate upkeep of social infrastructure after the cessation of the Bank's support, implying that the responsibility of upkeep may fall on communities that lack the resources to do so.

On balance, the evaluation rated the **sustainability** of the cluster projects as **satisfactory**, while noting strong differences across projects.

Additionality of non-sovereign operations

The Bank's presence was considered to have added value to the five non-sovereign operations assessed. For example, due to the Bank's credit ratings and track record, its involvement in projects allayed concerns around political risk, particularly in frontier markets that might otherwise not have attracted commercial capital.

Lessons

The following lessons were identified by the evaluation:

Lesson 1: Enhancing Project Performance

The Bank's technical support and close supervision of projects involving partner executing agencies with limited implementation capacity can significantly enhance project performance. Project assessments revealed varying levels of institutional capacity and strength among executing agencies. Assessing and addressing the institutional capacity of executing agencies at the outset of projects is key to enhancing project performance. Where capacity is found to be deficient, including Bank technical support and

close supervision as a component of the project interventions, can help to address the challenges and enhance project performance.

Lesson 2: Demonstrating Additionality

Given the capital-intensive nature of investments in the railway and aviation sub-sectors, effectively demonstrating the Bank's additionality is paramount. This ensures that the limited resources of the Bank are allocated appropriately and allows the Bank to highlight its unique contributions beyond what is available in the market, ensuring it does not crowd out the private sector and that it complements other financiers. For example, the Bank can play a crucial role by offering long-term financing for high-risk, large-scale infrastructure projects that commercial banks typically avoid. Furthermore, the AfDB can provide essential non-financial support, such as technical expertise, policy advice, and capacity building, to enhance project design and implementation.

Lesson 3: Financial Sustainability

A well-thought-out long-term financial sustainability plan for capital-intensive operations such as railways and aviation projects can enhance efforts to implement appropriate and adequate maintenance programs that extend beyond the Bank's involvement.

Lesson 4: Robust M&E Systems

A robustly designed and implemented monitoring and evaluation (M&E) system is crucial for the successful

delivery of interventions, as it helps track progress and identify areas needing improvement. However, project assessments found that while logical frameworks were generally well-articulated, the quality, adequacy, design, implementation and use of M&E systems were often insufficient. Key issues included inadequate risk identification, insufficient monitoring of implementation progress, and poorly planned ancillary components. Additionally, the lack of well-defined logic models, measurable indicators, and baselines limited the evaluability of projects. Ensuring that the M&E system avoids these shortcomings can promote timely data analysis for accountability, management, and learning.

Lesson 5: Environmental and Social Safeguards

Implementing adequate environmental, social, and operational safeguards for investments in physical infrastructure can prevent potential environmental degradation and protect the livelihoods of beneficiary communities. The evaluated cluster projects were found to have largely created opportunities for complementary components and separate value-addition projects that promote environmental and social causes in the affected areas. By doing so, the Bank added value not only as a financier but also as a promoter of sustainable development. The Bank's reputation helped allay concerns around environmental, social, and political risks held by other development partners, encouraging their buy-in. However, compliance with the safeguards by the borrower may be unlikely if this obligation is not defined in the borrowing terms. ■



Introduction

This report presents the findings of a cluster evaluation of 16 transport projects in the rail and aviation sub-sectors financed by the African Development Bank Group (AfDB or “the Bank”) between 2012 and 2023. It builds on a previous [cluster evaluation of road and port projects supported by the Bank during the 2012–2019 period](#).

The current section outlines the purpose and scope of the evaluation, as well as the methodological approach used for data collection, analysis, and synthesis of findings. The following section provides an overview of the Bank’s broader transport project portfolio. This is followed by a dedicated analysis of evaluation results, structured according to internationally recognized evaluation criteria: relevance, coherence, effectiveness, efficiency, and sustainability.

The penultimate section highlights key cross-cutting issues, specifically gender, climate change, and state fragility. The final section draws conclusions from the evaluation findings and presents strategic and operational lessons to inform future Bank operations.

Rationale, Purpose and Scope of the Evaluation

Independent Development Evaluation (IDEV)’s 2022–2024 Work Program, approved by the AfDB Board of Directors in December 2021, included a cluster evaluation of Bank-supported rail and aviation projects in Regional Member Countries (RMCs) covering the period 2012–2023, to complement IDEV’s previously conducted [cluster evaluation of road and port projects](#) (2021) and to provide specific inputs to the planned sector [evaluation of the Bank’s support for the transport sector 2012–2023](#) – the two cluster evaluations together cover

the four transport sub-sectors that the Bank has defined.

The evaluation examined a cluster of 16 AfDB-supported rail and aviation transport projects (the cluster projects, Table 1) and assessed their relevance, coherence, efficiency, effectiveness, and sustainability. It identified pertinent lessons drawn from the Bank’s engagements to inform the planning, design, management and implementation of future rail and aviation projects.

Serving as a building block of the broader transport sector evaluation, the main objective of this cluster evaluation was to generate evidence-based findings and draw pertinent lessons on what worked, what did not work and why, to inform future strategic and operational directions for the Bank’s assistance in the rail and aviation sub-sectors and the transport sector in general. The specific objectives of the evaluation were to:

- Assess to what extent operations in the transport sector were strategically aligned with the Bank’s TYS (2012–2023), and the needs, policies and priorities of RMCs.
- Assess to what extent recent operations incorporate innovative approaches and lessons learned from previous evaluations, and whether they were strategically aligned with the High 5s.
- Assess whether interventions were achieving intended results for the direct beneficiaries in terms of regional integration, connectivity, affordability, safety, and transport sector governance.
- Assess the extent to which the results achieved were sustainable; and

- Identify lessons to inform the Bank's future transport sector policies and/or strategies.

The evaluation covered projects approved during the period 2012–2023, as well as projects approved before but completed during the evaluation period. In addition, it covered the rail and aviation sub-sectors with respect to investment loans and grants, guarantees, equity participations, technical assistance (TA), capacity building, and analytical and advisory services towards the development of RMCs' specific rail and air transport sub-sectors, as well as the Bank's support to major regional rail and aviation strategies and programs.

The evaluation covered the primary modes of moving passengers and freight associated with the two sub-sectors including:

- Rail transport (infrastructure, rolling stock and equipment, and passenger and freight transport services).
- Air transport (airport infrastructure, aeronautical services, aircraft, and air transport services).

Projects were selected purposively to identify good practices and draw lessons accordingly. Selection criteria included: (i) contribution to Bank's High 5s, i.e., Integrate Africa and Industrialize Africa; (ii) regional representation; (iii) country categorization (Middle-income Country, Low-income Country, transition states, etc.); (iv) type of financing (African Development Bank, African Development Fund, Nigeria Trust Fund), and (v) scale/size of the operation.

The cluster evaluation was guided by the following questions:

- EQ1. To what extent did the objectives and design of the Bank's interventions respond to beneficiaries' global, country, and partner/institution needs?
- EQ2. To what extent did other Bank interventions (particularly policies) support or undermine the Bank's interventions in rail and air transport sub-sectors and vice versa? And to what extent were the

AfDB's interventions complementary, harmonized and coordinated with other development partners' support to RMCs?

- EQ3. How effective was the Bank in achieving its policy and strategic objectives and results (outputs and outcomes) regarding the two sub-sectors?
- EQ4. How efficient (timeliness and resource use) was the Bank in delivering its support to the two sub-sectors in RMCs?
- EQ5. How sustainable are the development results of the Bank's interventions?

Approach, Methods and Limitations

The evaluation adopted an approach encompassing a group of projects with common characteristics but implemented in different contexts (project cluster) to strengthen the external validity of the findings. A strong engagement process with key stakeholders was undertaken to increase the usefulness and utility of the evaluative evidence generated by this cluster evaluation.

The methodology adopted was based on a transport sector Theory of Change (ToC) presented in Annex 1, which provided a framework to direct the evaluation components. Based on the ToC, a project assessment rating grid — presented in Annex 2 — was developed to outline the specific judgement criteria against which each of the projects should be scored. These judgment criteria were designed to allow the evaluator to draw conclusions about five dimensions of project performance, namely relevance, coherence, effectiveness, efficiency and sustainability. In addition, non-sovereign operations were reviewed in terms of their additionality.

The evaluation brought together findings from:

- A portfolio review of the Bank's investments in the transport sector made between 2012 and 2023.

Table 1: Projects included in the rail and aviation cluster evaluation

Desk-based reviews	Country	Sub-sector	Funding source	Funding amount (UA m)	Funding type	Year of approval
Sharm El-Sheikh Airport Development Project	Egypt	Aviation	Sovereign	1.2	Grant	2015
Jomo Kenyatta International Airport Airfield Expansion	Kenya	Aviation	Sovereign	119.7	Loan	2017
Nacala Rail Corridor and Port Project & Value Addition Project	Multi-national: Mozambique - Malawi	Rail	Non-Sovereign & Sovereign	82.3	Loan	2015
Support for the Air Transport Sectors of West & Central Africa	Multi-national	Aviation	Sovereign	8.0	Grant	2015
Support to AUC/AFCAC on Single African Air Transport Market (SAATM)	Multi-national	Aviation	Sovereign	5.0	Grant	2020
Creation of a Leasing Platform for African Airlines	Multi-national	Aviation	Non-Sovereign	0.3	Grant	2020
Feasibility Study on the Extension of the Cameroon-Chad Railway Line	Multi-national: Cameroon - Chad	Rail	Sovereign	2.0	Loan	2017
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	Multi-national: Ethiopia - Sudan	Rail	Sovereign	1.5	Grant	2019
Air Côte d'Ivoire Modernization and Expansion Program	Côte d'Ivoire	Aviation	Sovereign	41.9	Loan	2017
Ghana Airports Company Limited (GACL) Capital Investment Programme	Ghana	Aviation	Non-Sovereign	89.8	Loan	2015
Corporate Loan to Ethiopian Airlines	Ethiopia	Aviation	Non-Sovereign	95.4	Loan	2016
Morocco - Railway Infrastructure Reinforcement Project	Morocco	Rail	Sovereign	84.0	Loan	2016
Democratic Republic of Congo - Priority Air Safety Project Phase II	Democratic Republic of Congo	Aviation	Sovereign	158.0	Loan	001 in 2010 / 003 in 2018
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	Senegal	Rail	Sovereign	149.8	Loan	2017
South Africa - Corporate Loan to Transnet	South Africa	Rail	Non-Sovereign	139.4	Loan	2014
Tunisia - Rail Infrastructure Modernization (II)	Tunisia	Rail	Sovereign	48.2	Loan	2003

Source: IDEV, based on Bank internal databases (as of November 14, 2024).

- In-depth case studies of eight of the Bank's rail and aviation projects approved and/or completed between 2012 and 2023.
- A desk-based review of a further eight² of the Bank's rail- and aviation projects approved and/or completed between 2012 and 2023.

The projects chosen for the in-depth and desktop assessments represent a mix of sovereign and non-sovereign interventions and — with representation across the continent's five regions — featuring a wide geographical reach. With regional integration at the core of several of the Bank's policies and strategies, six multi-national projects were selected for the evaluation. The projects selected for assessment

were all either completed, ongoing but nearing completion, or cancelled during implementation.

The evaluation combined both qualitative and quantitative methods (mixed methods). To feed into the purpose of the overall transport sector evaluation, the cluster evaluation aimed to determine the success or failure (accountability) of each individual project in the cluster, as well as to learn from the experience across the project cluster. The project assessments used a theory-based approach. This provided the basis for assessing results both at the individual project and project cluster levels. The evaluation also used a common data collection protocol to collect both quantitative and qualitative data on the performance of each selected project, with a standard results measurement framework for each sub-sector. The data was generated from multiple sources and collection methods including:

- **Document review:** Relevant Bank policy and strategy documents, project documents (loan agreements; appraisal, implementation and completion reports), as well project audit reports, were reviewed to assess the extent to which the projects have demonstrated efficiency in the use of resources.
- **Stakeholder interviews:** Key stakeholders engaged include Bank senior management, country and regional offices, operations teams, and project beneficiaries in RMCs as part of the project case studies.
- **Field visits:** The evaluation conducted field visits (where feasible) to selected project sites and held discussions with local officials, non-governmental organizations, development partners and a sample of the project beneficiaries.

Each category of data was analyzed using mainly descriptive statistics. Comparative analysis was also done at the indicator levels using baselines, targets and actual results. These lines of evidence were used iteratively and complementarily to cross-

reference the results of the analysis and to overcome any evaluation limitations.

The evaluation used a four-point rating scale³ for each evaluation criterion. The ratings on each criterion are presented as follows in the report:

	Highly Satisfactory
	Satisfactory
	Partly Unsatisfactory
	Unsatisfactory
	Not available

Limitations

Even though triangulation of data sources was used to minimize limitations to the extent possible, some limitations were observed including:

- **Data availability and consistency.** At the time of the review, the projects were at different stages of implementation, with some already completed, some cancelled, and the remainder ongoing but nearing completion. In the case of ongoing or cancelled projects, project completion reports (PCRs) were naturally not available, meaning that the evaluators relied on sources such as implementation progress and results reports (IPRs) and/or stakeholder interviews to arrive at conclusions around implementation performance.
- **Although the same rating grid was used to assess the performance of all projects, it should be noted that some of the individual judgment criteria did not apply to all projects.** For example, those that were cancelled prior to implementation, such as the Sharm El-Sheikh Airport Development Project, were not assessed in terms of the judgment criteria linked to implementation (i.e., those that contributed to the effectiveness and efficiency scores).
- **Some expected documentation was not readily available for all projects.** For example, neither IPRs nor back-to-Office Reports (BTORs) for

the feasibility study on the extension of the Cameroon-Chad Railway Project were availed, nor were its results measures published on the Bank's data portal. This precluded analysis relating to the evaluation criteria linked most closely to implementation (i.e., effectiveness and efficiency).

- Some case studies relied on different documents to draw conclusions around the same evaluation criteria.

For example, in the case of the Nacala Rail Corridor and Port Project, BTORs and the limited data available on the Bank's data portal were used in lieu of IPRs to draw conclusions around effectiveness.

While these limitations are worth noting, the information provided was generally sufficient to assess most projects' performance in all five dimensions of the evaluation criteria. ■





Overview of the AfDB Transport Sector Portfolio

This section provides an overview of the Bank's transport sector portfolio trends from 2012 to 2023. It also reviews the Bank's financial commitments to the transport sector in general, with a specific focus on its funding for the rail and aviation sub-sectors during this period.

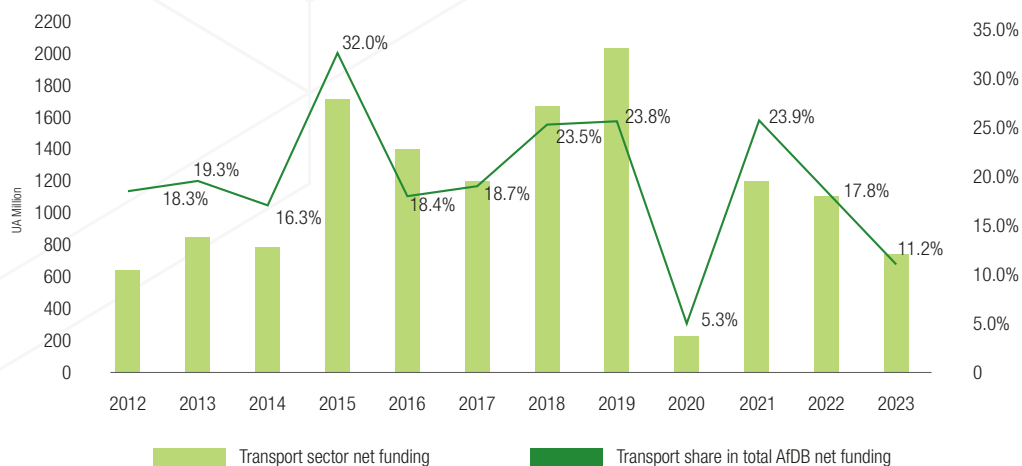
The main database used for the portfolio analysis is the Bank's Systems Applications and Products in Data Processing (SAP). The crossing of the SAP data with the Bank's Infrastructure and Urban Development Department and Non-Sovereign Operations and Private Sector Support Department databases made it possible to validate the data and collect some missing information. An analysis of the project duration and of the time-lapse between project approval and the first disbursement could

only be performed for the projects for which this information was available.

Transport Sector Portfolio Trends

The Bank's financial commitments in the transport sector during the period 2012-2023 were mainly driven by the road sub-sector, which drew 75.2% of total net financing and 69.7% of the number of projects. The rail and aviation sub-sectors were among the least financed by the Bank. The period 2012-2023 recorded a significant fluctuation in funding for transport projects, both in terms of value and as a share of the Bank's overall portfolio (Figure 1). A sharp increase in transport funding was recorded in 2015, resulting from the

Figure 1: Transport project funding over time - key trends (2012-2023)



Source: Calculated by IDEV, based on Bank internal databases (as of November 14, 2024).

Bank Group's prioritization of infrastructure as part of its Ten-Year Strategy (TYS) 2013-2022. The peak of transport financing (in volume terms) was reached in 2019, with a total of UA 2 billion. However, this was short-lived due to the COVID-19 pandemic, leading to a dramatic drop in funding to UA 244 million in 2020, as projects were restructured, and resources were reallocated to address immediate needs of RMCs. While transport funding levels recovered to UA 1.2 billion in 2021, they dropped again in 2022 and 2023. This is illustrated in Figure 1.

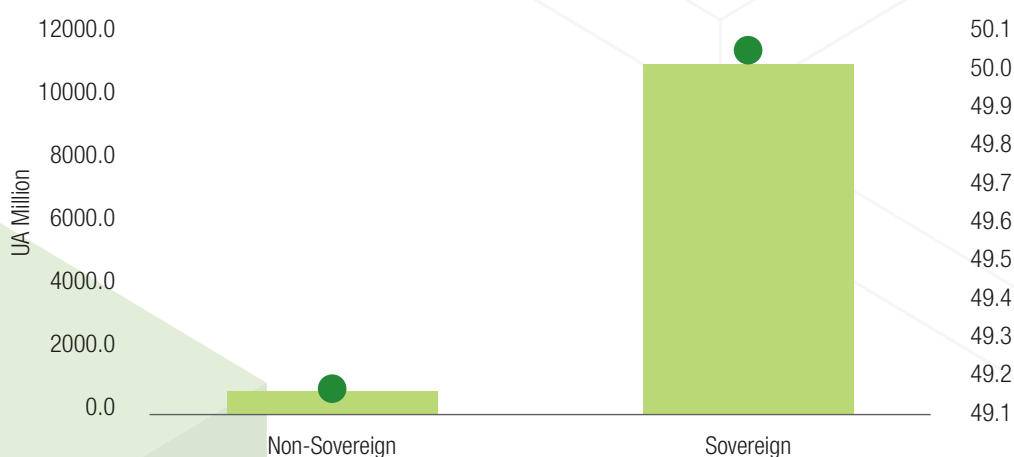
The Bank Group's support was significant vis-à-vis the amount of capital requested by RMCs. From 2012 to 2023, the AfDB's net financing amounted to about UA 14 billion for 258 transport interventions across its RMCs, recording a two-fold increase in funding compared to the 2000-2011 period (UA 7 billion). Although transport projects were fewer in number than those in the multi-sector, agriculture, and energy sectors, they accounted for the second-largest share of the Bank's total commitments by value, underscoring the capital-intensive nature of

large-scale transport investments. Of the total UA 14 billion pledged, UA 12 billion, or 85%, was allocated to state-guaranteed transport projects, highlighting the Bank's strong emphasis on government-backed initiatives, as illustrated in Figure 2.

AfDB's Financing to the Rail and Aviation Sub-sectors

Between 2012 and 2023, the AfDB financed 11 rail and 25 aviation projects with total net funding of UA 1,145.25 million, specifically, UA 389.11 million and UA 756.14 respectively. While the aviation sub-sector received more projects, the average funding per project was higher for rail (UA 35.4 million) compared to aviation (UA 30.3 million). The rail and aviation sub-sectors accounted for only 2.8% and 5.4%, respectively of the Bank's total transport sector funding, highlighting their relatively low prioritization compared to other sub-sectors like roads. A summary of the data is presented in Table 2.

Figure 2: Bank's commitments to Sovereign & Non-Sovereign Operations in the transport sector (2012-2023)



Source: Calculated by IDEV, based on Bank internal databases (as of November 14, 2024).

Table 2: Bank net funding, number of projects in rail and aviation sub-sectors (2012-2023)

Sub-sector	Total of net funding (UA Million)	Share in total net funding (%)	Number of projects	Share in total number of projects (%)	Average net funding per project (UA Million)
Air Transport	756.14	5.4	25	9.7	30.25
Rail Transport	389.11	2.8	11	4.3	35.37
Total	1,145.35	8.2	36	14.0	

Source: Calculated by IDEV, based on Bank internal databases (as of November 14, 2024).

The Bank's financial response to RMCs' needs in these sub-sectors was varied in terms of approved amounts. In both the rail and aviation sub-sectors, 100% of the requested amounts (UA 389.11 million & UA 756.14 million) were approved.

Over 90.1% (10 out of 11) of AfDB operations in the rail sub-sector were public sector driven, while in the aviation sub-sector 92% were sovereign operations. This reflects the scale and complexity of rail and aviation infrastructure, which typically requires large public investment and coordination. Despite the dominance of public sector financing, private sector participation has received

some interest and traction in the two sub-sectors. However, the aviation sub-sector has shown a greater ability to attract private investment.

The rail sub-sector, although having fewer projects (11 projects), demonstrated a more significant focus on infrastructure investment, with 98% of projects dedicated to it. The aviation sub-sector (25 projects), while similarly prioritizing infrastructure, also invested in advisory services and technical assistance. Both sub-sectors exhibited significant scope for further investment, given the continent's need for integrated transport networks to boost economic growth and regional trade. ■



Project Cluster Performance

This section presents a summary of the findings and ratings of the 16 project assessments using the international evaluation criteria. The process also benefited from the analysis of the Bank's transport sector portfolio, and desk and in-depth analysis of case study projects.

Relevance

Relevance measures the extent to which the development objectives of a project are consistent with the beneficiary needs, the country's development or policy priorities and strategy, the Bank's Country Strategy Paper (CSP) and the applicable Bank sector and regional strategies. The evaluation of relevance also considers the quality and appropriateness of the intervention's design — both from a technical perspective and a planning perspective.

Overall, the relevance of the cluster projects was assessed to be satisfactory (Figure 3), with

all 16 projects receiving scores of either 3 or 4 for the criterion. The cluster projects were generally found to be suitably aligned with the relevant strategies of the Bank, as well as the development strategies of the beneficiary RMCs. While most of the projects were viewed satisfactorily in terms of their design, some showed weakness in this regard.

Alignment

The cluster projects' objectives were assessed to be fully aligned with relevant strategies, policies and priorities of the Bank and other relevant entities. This included the Bank's Ten-Year Strategy (2013-2022), Country Strategy Papers (CSPs), Regional Integration Strategy Papers (RISPs), and High 5s, as well as the United Nation's Sustainable Development Goals (SDGs) and the African Union's (AU's) Agenda 2063. The favourable performance in this respect suggests that the Bank's selection criteria ensure that the transport projects it

Figure 3: Summative project ratings – Relevance

Creation of a Leasing Platform for African Airlines	4
Corporate Loan to Ethiopian Airlines	4
Morocco - Railway Infrastructure Reinforcement Project	4
Tunisia - Infrastructure Modernization (II)	4
Sharm El-Sheikh Airport Development Project	3
Jomo Kenyatta International Airport Airfield Expansion	3
Nacala Rail Corridor and Port Project & Value Addition Project	3
Support for the Air Transport Sectors of West & Central Africa	3
Support to AUC/AFCAC on Single African Air Transport Market	3
Feasibility Study on the Extension of the Cameroon-Chad Railway	3
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	3
Air Côte d'Ivoire Modernization and Expansion Program	3
Ghana Airports Company Limited (GACL) Capital Investment Programme	3
Democratic Republic of Congo-Priority Air Safety Project Phase II	3
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	3
South Africa-Corporate Loan to Transnet	3

funds are — at the conceptual level — responsive to the specific needs of RMCs and contribute to the broader vision that the Bank has for socio-economic development on the continent.

The evaluation found that the objectives of the evaluated cluster projects typically aligned with at least two of the five operational priorities of the Bank's TYS 2013-2022, with infrastructure development and regional economic integration being the most common. The Bank's TYS was built upon the overarching objectives of ensuring more inclusive growth and encouraging a transition to sustainable green growth on the continent through the financial and operational support of five sectoral priorities, namely infrastructure development, regional economic integration, private sector development, governance and accountability, and skills and technology. In addition, some cluster projects involved the private sector, either as financing, delivery partners or beneficiaries. Projects with capacity-building initiatives, either delivered as core project components or ancillary components, supported the “skills and technology” priority – for example, the Support to AUC/AFCAC on the Single African Air Transport Market (SAATM).

Moreover, the evaluation found the cluster projects to have full alignment with the Bank's five post-2015 operational priority areas (the High 5s), with the latter three priorities being referenced frequently in the project appraisal documents. Building on its TYS, the Bank set out five development priorities known as the High 5s in 2015, namely Light up and Power Africa, Feed Africa, Industrialize Africa, Integrate Africa, and Improve the Quality of Life for the People of Africa. Complementary initiatives that supported the agriculture sector and facilitated access to markets, such as the Nacala Rail Corridor and Port Value Addition Project, allowed the Bank to leverage its investments in physical infrastructure to contribute to the Feed Africa priority.

Additionally, cluster projects were found to demonstrate clear alignment with the strategic

intent of the Bank's RISPs, having a strong focus on the development of regional transport infrastructure, improved connections with existing regional networks, and supporting policy-based initiatives to foster regional cooperation. The Bank's CSPs and RISPs outline its strategic intent and priorities for RMCs' development in an integrated manner. Since the development of rail and aviation transport infrastructure is central to most RISPs, the relevant cluster projects' objectives shared obvious alignment with their respective nations' RISPs. The Bank's multinational investments in the two sub-sectors consisted of trans-border international airports and multinational rail. For example, the Bank's interventions in Southern Africa between 2012 and 2023 were found broadly consistent with the goals set out in the two RISPs covering the period — indeed, regional infrastructure forms a pillar of the strategy in each of these RISPs, with an emphasis on transport. The current RISP for Southern Africa for 2020-2026 also emphasizes cross-border rail links. It was found that the Bank is supporting rail projects going forward, targeting the construction of 1,500km of rehabilitated and modernized cross-border railway lines by 2026, as part of its priority area on infrastructure connectivity. In Eastern and Southern Africa, the COMESA Airspace Integration Project led to member states agreeing to a phased approach to airspace integration.

Although the aviation cluster projects were found to have clear benefits for integration, safety and broader socio-economic development, the expected increase in air traffic volumes due to their implementation is likely to have associated impacts on greenhouse gas (GHG) emissions. While this was typically acknowledged in project documentation, it was not always clear whether mitigation methods had intentionally been put in place to fully offset this effect.

Quality of project design

The evaluation found that all cluster projects had cogent and well-articulated logical frameworks. However, some challenges related to project

design were also observed, with consequences for the relative success and timeliness of project delivery. These included shortcomings relating to project objectives, risk identification and mitigation, integration of lessons learned from previous projects (whether Bank-financed or otherwise) and monitoring of implementation progress. Although projects' ancillary components typically had clear complementarity with their associated core components, their planning was generally found less comprehensive, contributing to delivery shortcomings. **Issues related to inadequacy and choice of outcomes and the monitoring thereof were highlighted in multiple assessments.** Even though the cluster projects included results-based management arrangements at the design stage, the design and use of monitoring and evaluation (M&E) were found inadequate. For instance, project implementation arrangements lacked well-defined logic models or frameworks for assessing outcomes and impacts. In some cases, there were no monitoring plans with measurable indicators and baselines at appraisal, thereby limiting the evaluability of the projects at midline and endline. Moreover, it was found that risk mitigation measures and the integration of lessons learned from previous interventions into projects' designs were often insufficient. Examples include the feasibility studies on the extension of the Cameroon-Chad Railway and on the Ethiopia-Sudan Standard Gauge Railway. Indeed, although the studies' designs incorporated lessons learnt from previous projects, as well as mitigation measures to avoid delays in delivery, it was largely unclear whether the studies' components considered the potential issues around operation and maintenance, or around the competitive advantages or disadvantages of rail transport on the continent.

Desk review of the Project Appraisal Reports (PARs) of the cluster projects found that in some projects,⁴ the links between the intervention and the selected impact indicators were noted as being “tenuous and inherently exposed to external influence”, undermining efforts to monitor both the implementation progress and

the projects' associated contribution towards achieving their core purpose. In addition, monitoring of progress was flagged as a concern in the Support for the Air Transport Sectors of West & Central Africa Project. Relatedly, inconsistencies surrounding progress and objective monitoring were observed across projects, with obvious differences in the quality and type of reporting made available for review.

The review of the cluster projects' PARs, PCRs and XSRs found that the quality of cluster projects' designs was influenced by a number of factors including: (i) comprehensive feasibility studies- adequacy of data with realistic forecasts and high quality geological and hydrological study or soil survey for a good siting of infrastructure projects; (ii) quick expedition of expropriation and land acquisition procedures at appraisal; readiness assessment of the project implementation at appraisal including sensitization, training, and education of the rural population as well as the socio-cultural acceptability at a local level; (iii) institutional study on cost recovery and affordability, particularly for users of rail and aviation services; (iv) environmental and social impact assessments of the projects, and (v) strong linkages between project objectives and selection of performance measurement indicators. For example, in the case of the Sharm El-Sheikh Airport Development Project, details around the implementation approach for a proposed Centre of Excellence were notably vague, with no information relating to specific implementation arrangements and timelines provided. Similarly, in the case of the SAATM, there was a disconnect between performance indicators and the project's objectives, as presented in Box 1.

Coherence

Coherence assessed the complementarity and compatibility of the project cluster with other interventions supported by the AfDB (i.e., internal coherence), the government and other development partners (i.e., external coherence).

Box 1: Issues relating to performance indicator selection - Support to AUC/AFCAC on Single African Air Transport Market

While the project's broad objectives supported the realization of the overarching visions of both the AfDB and the AU, some of the indicators selected to measure progress and success were found, to a large degree, to be exposed to external factors. As a result, the possibility of a disconnect between these indicators and the project's relative success was apparent. For example, the number of air transport passengers and the proportion of intra-African routes operated by more than one airline were both linked to the establishment of the Single African Air Transport Market (SAATM) but were also intrinsically and strongly linked to external factors, such as market conditions and the existence of adequate physical infrastructure. The onset of the COVID-19 pandemic, which saw passenger numbers decline considerably despite the African Civil Aviation Commission (AFCAC) achieving simultaneous progress on other core project objectives, demonstrated this issue.

Setting targets that are largely influenced by external factors can detract from the Bank's monitoring and evaluation efforts. This makes it difficult to assess the progress and quality of work, and in turn hinders the ability to identify corrective measures to challenges faced. This was highlighted in the 2022 IPR, which indicated that significant progress was made on the ground, despite the chosen quantitative indicators suggesting underwhelming performance.

Source: Desk assessment of P-Z1-DAO-015 Multinational - Institutional Support to the African Civil Aviation Commission for the Implementation of the Single African Air Transport Market.

Both internal and external coherence were found satisfactory for all the projects in the cluster. The summative project ratings were derived from the average of project scores for internal and external coherence (Figure 4). Overall, the cluster projects received the highest ratings for coherence compared to other evaluation criteria.

The evaluation found all the cluster projects to have satisfactorily demonstrated synergies and interlinkages with other Bank interventions, as well as with the initiatives of other development partners. Internal coherence within both the rail and aviation sub-sectors was found to be strong. All the cluster projects were found to have robust synergies and interlinkages with other projects in the indicative operational program of the CSPs and/or the existing portfolio in the case study countries, which highlighted development objectives around regional integration with a strong connection to transportation. Four projects were found to be follow-up phases of previous Bank-funded operations. These include Tunisia - Rail Infrastructure Modernization (I), South Africa - Corporate Loan to Transnet (I), and the Democratic Republic of Congo - Priority Air Safety Project Phase I). This suggests that the Bank's individual projects tend to avoid duplication of effort and generally contribute to broader synergies at local, national or regional levels.

Some incongruence was, however, noted in the aviation sub-sector between projects that supported the acquisition of aircraft by airlines and those which sought to promote aircraft leasing among the same target beneficiaries. Although most aviation projects related to institutional capacity building and airport infrastructure development demonstrated coherence and complementary, the synergies between these projects and the Bank's financial support of airlines were less evident. The evaluation also noted that, although complementarity might exist between the Bank's transport projects, there was often little evidence of this being achieved as the result of a deliberate plan. Such complementarity between the Bank's interventions has both direct benefits in the transport sector, and for the economic growth and development of the RMCs.

With one exception, the cluster projects' external coherence was rated as either satisfactory or highly satisfactory. For example, the Project for Support to the Air Transport Sectors of West and Central Africa (PASTA-CO) was found to have strong coherence with the European Union-Africa Safety in Aviation Project to build the capacity of Regional Safety Oversight Organizations in Africa, by developing harmonized regulations, common oversight processes, and providing training regionally and throughout the continent, which contributes to

Figure 4: Summative project ratings - Coherence

Corporate Loan to Ethiopian Airlines	
Feasibility Study on the Extension of the Cameroon-Chad Railway	
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	
Morocco-Railway Infrastructure Reinforcement Project	
Nacala Rail Corridor and Port Project & Value Addition Project	
Support for the Air Transport Sectors of West & Central Africa	
Support to AUC/AFCAC on Single African Air Transport Market	
Creation of a Leasing Platform for African Airlines	
Tunisia Rail Infrastructure Modernization (II)	
Air Côte d'Ivoire Modernization and Expansion Program	
Jomo Kenyatta International Airport Airfield Expansion	
Sharm El-Sheikh Airport Development Project	
Ghana Airports Company Limited (GACL) Capital Investment Programme	
Democratic Republic of Congo - Priority Air Safety Project Phase II	
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	
South Africa Corporate Loan to Transnet	

regional integration. This is also in alignment with the African Civil Aviation Policy established by the African Union, which seeks to harmonize aviation policies, regulations and procedures, and to integrate aviation systems. Similar coherence was found with the International Air Transport Association's 2016 call to African governments to ensure the development of aviation at national and continental levels, to foster economic growth through the establishment of regulations focused on safety and improved connectivity.

Conversely, the Jomo Kenyatta International Airport Airfield Expansion Project received an unsatisfactory rating for external coherence due to there being little evidence of coordination between the Bank and other development partners involved in separate ongoing initiatives at the airport, as well as an apparent lack of an overarching development strategy to direct distinct, but potentially complementary initiatives at the airport.

Box 2 contains an extract from the assessment of the *Creation of a Leasing Platform for African Airlines* Project that gives an example of mixed coherence.

Effectiveness

The assessment of effectiveness considered the extent to which the cluster projects achieved their stated objectives (results), as reflected in their outputs and outcomes. The analysis was based on comparisons between the actual and planned results, as set out in the projects' logical frameworks and measured using associated quantitative indicators. The summative project ratings (Figure 5) were the average of project scores for outputs and outcomes.

Overall, most evaluated cluster projects were assessed to have satisfactorily delivered their results. However, two ongoing projects — the Support for the Air Transport Sectors of West & Central Africa and Support to AUC/AFCAC on Single African Air Transport Market — were yet to satisfactorily achieve their stated outputs and outcomes, despite being well beyond their planned completion dates.

Achievement of overall objectives

The evaluation found the projects' development of physical infrastructure schemes to have

Box 2: Mixed project coherence - Creation of a Leasing Platform for African Airlines

Given that promoting regional integration is a core facet of the Bank's High 5 priorities, many of its investments are designed with the intention of fostering regional connectivity as a central objective. In the context of the aviation industry, the Bank's support is guided by its Framework and Guidelines for Support to the Aviation Sector, which aims to overcome entry barriers and poor connectivity to promote competition and financial sustainability. The creation of an airline leasing platform with synergies and interlinkages with many of the Bank's ongoing initiatives in the aviation and broader transport sector implies a high degree of internal coherence. Clear complementarity exists between the project and the Bank's investments in capacity-building initiatives (e.g., Project for Support to the Air Transport Sectors of West and Central Africa and Institutional Support to the AFCAC for the Implementation of the SAATM) and airports (e.g., Rabat-Salé Airport Modernization and Extension Project). Linkages between the airline leasing platform and the Bank's support for airlines (e.g., Air Côte d'Ivoire Modernization and Expansion Program and Corporate Loan to Ethiopian Airlines) are arguably more tenuous, as these projects have involved supporting the acquisition of new aircrafts as opposed to promoting aircraft leasing. While it might be reasoned that Ethiopian Airlines does not fall within the scope of the leasing platform project's target beneficiaries due to its large size and long-established presence, Air Côte d'Ivoire's small fleet and relative youth make it a textbook candidate for the leasing platform. Although the acquisition of these aircrafts does not inherently preclude further expansion by means of leasing, it could be argued that it still bears an opportunity cost in the broader context of the Bank's efforts to promote leasing.

Source: Project assessment of P-Z1-DA0-019 - Creation of a Leasing Platform for African Airlines.

direct benefits for beneficiaries in terms of connectivity, operational efficiency and safety.

The effectiveness rating considered individual project performance in achieving their stated outputs and expected development outcomes. Ten out of twelve (10/12) projects satisfactorily delivered their output targets, with an additional two (2/12) achieving a partly unsatisfactory rating. However, four projects were not rated due to either cancellation⁵ or unavailability of information.⁶ Relative to outcomes, six projects satisfactorily achieved their targets, three being rated partly unsatisfactory and two rated unsatisfactory. The few satisfactory ratings on project outcomes were driven by the achievement of key outcomes, including improved transport efficiency, reduced air accidents, increased connectivity, etc. A summary of individual project ratings for effectiveness is provided in Figure 5.

Achievement of projects' outputs

An assessment of the achievement of the project outputs was performed by comparing the level of achievement of the actual outputs with the targets set at project appraisal within each project component.

The evaluation found that while the projects' performance in terms of relevance and coherence - which principally relate to their pre-implementation stages - was generally satisfactory, they had mixed levels of success during implementation, as reflected in their attainment of target outputs. Projects focusing on the delivery of physical transportation infrastructure and/or train units appear to have either delivered in full or, for the most part, met their output objectives (Table 3). Although most of the projects recorded implementation delays, they generally delivered their core components to appropriate design specifications and standards, with operating performance achieving planned service levels. In contrast, the projects' ancillary components were only partly delivered - often towards the end of implementation schedules - or cancelled altogether, as was the case with the Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I, which did not achieve the majority of its ancillary component outputs. Similarly, the projects that centred around institutional capacity building had limited success in achieving their intended outputs, with capacity limitations of the executing agencies generally being a core

Figure 5: Summative project ratings - Effectiveness

Corporate Loan to Ethiopian Airlines	
Nacala Rail Corridor and Port Project & Value Addition Project	
Creation of a Leasing Platform for African Airlines	
Air Côte d'Ivoire Modernization and Expansion Program	
Ghana Airports Company Limited (GACL) Capital Investment Programme	
Morocco - Railway Infrastructure Reinforcement Project	
Democratic Republic of Congo - Priority Air Safety Project Phase II	
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	
South Africa Corporate Loan to Transnet	
Tunisia - Rail Infrastructure Modernisation (II)	
Support for the Air Transport Sectors of West & Central Africa	
Support to AUC/AFCAC on Single African Air Transport Market	
Feasibility Study on the Extension of the Cameroon-Chad Railway	N/A
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	N/A
Sharm El-Sheikh Airport Development Project	N/A
Jomo Kenyatta International Airport Airfield Expansion	N/A

constraint. Details of performance indicators and results achieved for each of the cluster projects are provided in Annex 3.

Of the 13 implemented⁷ cluster projects, eight projects⁸ received either highly satisfactory or satisfactory ratings for their output achievements (more than 75% delivery). The assessment highlighted that most output objectives had either already been met, exceeded, or were well on course to being achieved by their target years. However, three projects were rated partly unsatisfactory for their limited achievements (between 50% and 75%), and one project was rated unsatisfactory for achieving less than 50% of its output deliverables. **The evaluation found that infrastructure was typically developed to an appropriate degree of quality, complying with relevant quality and safety standards** (e.g., Tunisian Railway Infrastructure Modernization Project Phase III). It also found that some adjustments to project scopes were made during implementation, but these amendments were noted as being responsive to changing beneficiary needs and followed Bank approval process requirements. Table 3 presents a summary of the achievement of outputs.

Implementation challenges related to planned ancillary components were noted in several project assessments, explaining the partial delivery of project outputs. Table 4 lists the complementary outputs (ancillary components) achieved. In the case of the Air Côte d'Ivoire Modernization and Expansion Program, which received an unsatisfactory rating, it was found that the outputs related to aircraft acquisition, training and the business plan for Félix Houphouët-Boigny National Polytechnic Institute (INP-HB) were not fully implemented. In addition, the partial risk guarantee, aviation policy study and the financial audits were also not completed. Box 3 gives an additional illustration of the challenges in implementing an ancillary component in a capacity-building project.

Achievement of projects' development outcomes

The evaluated cluster projects were assessed on the attainment of the development outcomes identified at project appraisal. Direct project outcomes were generally related to improved domestic and international connectivity, efficiency, safety, and institutional capacity, as reflected in the ToC.

Table 3: Summary of project outputs achieved

Project Name	Number of targets achieved	Ratio of targets achieved (%)	Rating
Support for the Air Transport Sectors of West & Central Africa	11/18	61	Partly unsatisfactory
Support to AUC/AFCAC on Single African Air Transport Market	3/5	60	Partly unsatisfactory
Feasibility Study on the Extension of the Cameroon-Chad Railway	4	N/A	N/A
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	3/4	75	Satisfactory
Creation of a Leasing Platform for African Airlines	3/4	75	Satisfactory
Air Côte d'Ivoire Modernization and Expansion Program	2/5	40	Unsatisfactory
Ghana Airports Company Limited (GACL) Capital Investment Programme	2/2	100	Highly satisfactory
Corporate Loan to Ethiopian Airlines	3/3	100	Highly satisfactory
Morocco - Railway Infrastructure Reinforcement Project	7/8	87.5	Satisfactory
Democratic Republic of Congo - Priority Air Safety Project Phase II	22/22	100	Highly satisfactory
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	8/16	50	Partly unsatisfactory
South Africa - Corporate Loan to Transnet	1/1	100	Highly satisfactory
Tunisia - Rail Infrastructure Modernization (II)	9/12	75	Satisfactory

Source: Project assessments (data from project completion reports and site visits).

Despite the successful attainment of output results by most of the implemented cluster projects, the evaluation found a mixed picture relating to the attainment of outcomes. Aside from the two projects that did not have data for outcome assessments, one project achieved a highly satisfactory rating for its outcomes, five projects received satisfactory ratings, three projects got partly unsatisfactory ratings, and

two attained unsatisfactory ratings. Notwithstanding the poorer outcome ratings, overall, the development of physical infrastructure yielded direct benefits for beneficiaries in terms of connectivity and operational efficiency and, where relevant, safety improvements, which were achieved satisfactorily. Table 5 presents a summary of the cluster projects' achievement of outcomes.

Table 4: Delivery of ancillary components in selected projects

Project Name	Commentary on delivery of ancillary components
Aviation project	
Democratic Republic of Congo - Priority Air Safety Project Phase II	Sensitization of the population within the project impact area on HIV/AIDS achieved a completion rate of only 3%.
Rail projects	
Morocco - Railway Infrastructure Reinforcement Project	Non-implementation of the activities under the " <i>consolidation of gender achievements</i> " component was explained by the Moroccan National Railways Office's inability to conclude an agreement with local authorities towards the implementation of the planned activities.
Tunisia - Rail Infrastructure Modernization Project (II)	Improved access to services and jobs for women was intended to be achieved, however, there was little evidence to justify that benefits had necessarily been delivered to women.
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	The intended socio-economic infrastructure and facilities had not been completed and were at risk of being cancelled. Only one of the planned sports fields was nearing completion, and the women's centres were in early stages of development. The Agency for the Promotion of Investments and Major Works cited challenges with local contractors and delayed payments by funders, including AfDB, as primary reasons for late development. It also commented that the other planned initiatives (apart from the sports fields and women's centres) would likely be cancelled.

Source: Project assessments (data from project completion reports and site visits).

Box 3: Implementation challenges in capacity building projects - Support for the Air Transport Sectors of West & Central Africa Project

The provision of institutional support and various studies and diagnostics were slowed by delays in obtaining consultants' services and financing from the International Civil Aviation Organization (ICAO) for such activities, however the outputs had attained 80% completion by 2023. Activities related to the training of national inspectors in air safety, safety audits and Airport Excellence Program (APEX) reviews linked to airport certification, which were under the management of the ICAO and European Union Aviation Safety Agency, were experiencing significant delays as of 2024. Although inspector manuals and certifications were said to be granted frequently, the latest IPR (2023) showed no further evidence or numerical values to illustrate the extent of their delivery.

As of 2024, only 11 of the 20 targeted airports in the regions were compliant with international standards, and no effective local runway safety teams were implemented in each international airport, despite an ambitious target of 30 teams. None of the 10 operator certifications were delivered, nor were Aviation Security (AVSEC) pools set up. Furthermore, the issues in the project resulted from delays by the executing agencies in disbursing funding, which hindered the implementation of planned activities, particularly in the areas of training and certifications. This was in addition to the lack of coordination exhibited by the ICAO and the West African Economic and Monetary Union (WAEMU) in realizing capacity-building components.

Source: Assessment of P-Z1-DAO-010/011/012 Project for Support to the Air Transport Sectors of West and Central Africa (PASTA-CO).

Improved connectivity and transport efficiency gains

In-depth case studies found increased connectivity and transport efficiency gains in traffic volumes despite the limited availability of updated traffic data for several assessed cluster projects. In relation to this outcome, the Corporate Loan to Ethiopian Airlines to modernize and expand its fleet contributed to an increase in connectivity from 94 international and 21 domestic destinations in 2016 to 127 international and 22 domestic destinations in 2022. This resulted in an increase in passenger numbers from 719,972 in 2016 to 1,151,529 departing passengers in 2024. Similarly, the Bank's support to Air Côte d'Ivoire to augment its fleet also contributed to improved national and regional connectivity, with an increase from 3 domestic and 19 regional destinations in 2016 to 5 domestic and 20 regional destinations in 2024.

Connectivity was deemed to have improved at either local or regional levels, depending on the intended purpose of the infrastructure. Most of the cluster projects included the construction and/or upgrade of infrastructure with regional significance, such as airports and inter-city or cross-border rail systems, enhancing target populations' access to

regional social and economic opportunities and promoting businesses' integration into international supply chains. Among the projects reviewed, the Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I covered a relatively localized area of influence, however, its connection to Senegal's Blaise Diagne International Airport in the second phase of the project had a direct impact on Dakar's connectivity, although capacity constraints prevented this greenfield rail project from reaching its targeted passenger volume level of 115,000 from the current 77,000 (2024).

The desk analysis of the cluster projects (including PCRs, XSRs, and IPRs) revealed that the cluster projects had not contributed to increased volumes of cargo transport. For the rail projects, cargo transport volumes did not record a consistent increase over time, with some periods recording lower-than-expected volumes resulting from several external factors, including the impact generated by the COVID-19 pandemic on global trade. On the other hand, according to the XSR of the Corporate Loan to Transnet II, the volumes in the general freight business grew to a record 90.8 million tonnes (mt) per annum in 2018, relative to 88.1 mt per annum in the previous year. The total volume in 2017/18 was 226 mt. However, there was

Table 5: Summary of project outcomes achieved

Project Name	Number of Targets achieved	Ratio of targets achieved (%)	Rating
Support for the Air Transport Sectors of West & Central Africa	4/6	67	Partly unsatisfactory
Support to AUC/AFCAC on Single African Air Transport Market	2/3	67	Partly unsatisfactory
Feasibility Study on the Extension of the Cameroon-Chad Railway	3	N/A	N/A
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	3	N/A	N/A
Creation of a Leasing Platform for African Airlines	1/3	33	Unsatisfactory
Air Côte d'Ivoire Modernization and Expansion Program	2/5	40	Unsatisfactory
Ghana Airports Company Limited (GACL) Capital Investment Programme	3/5	60	Partly unsatisfactory
Corporate Loan to Ethiopian Airlines	5/6	83	Satisfactory
Morocco - Railway Infrastructure Reinforcement Project	3/4	75	Satisfactory
Democratic Republic of Congo - Priority Air Safety Project Phase II	5/6	83	Satisfactory
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	4/5	80	Satisfactory
South Africa - Corporate Loan to Transnet	1/1	100	Highly satisfactory
Tunisia - Rail Infrastructure Modernization Project (II)	5/6	83	Satisfactory

Source: Project assessments (data from project completion reports and site visits).

a steady decrease in rail volumes between 2017/18 and 2022/23, primarily attributed to a challenging operating environment and binding constraints including vandalism, reduced locomotive tractive effort availability and underinvestment in the rail network, along with a rolling stock backlog. Volumes had reduced to 149.5 mt, with the general freight business accounting for 49.7 mt. The total volume therefore reduced by approximately 33%, whereas the general freight business volume reduced by almost 50%.

The desk-review of cluster projects and field validations found that transport efficiency gains were recorded for both rail and aviation sub-sectors. In the rail sub-sector, the Tunisian Railway Infrastructure Modernization Project (Phase II) contributed to time gain improvements on the main lines, including the Tunis-Sousse-Sfax-Gabès (30 minutes), Tunis-Ghard (10 minutes), and Tunis-Radès (10 minutes by train / 35 minutes by car). Furthermore, the Tunisian Railway Infrastructure Modernization Project (Phase II) has led to scheduling improvements, with the percentage of trains arriving less than 15 minutes late increasing from 70% (2010) to 80% (2022). Additionally, the number of phosphate trains arriving more than 120 minutes

late decreased significantly from 91% in 2010 to 20% in 2022. Also, there has been a turnaround time of 1.7 trains/day (2024) compared to 1.0 trains/day (2010).

Similarly, for the Dakar-Diamniadio-AIBD Regional Express Train Project – Phase I, the average transit time in 2015 was 108 minutes (60 minutes of journey time and 48 minutes of waiting time per passenger) with a commercial speed of 20 km/h for the Petit Train de Banlieue (PTB). Upon completion of Phase I of the Regional Express Train (TER), transit times improved to 55 minutes (45 minutes journey time with 10-minute headways until 9:00pm from Monday to Saturday, and 20-minute headways as of 10:00pm and on Sundays) and a commercial speed of 50 km/h, with a maximum speed of 160 km/h. Congestion was reduced through better utilization of public transport. In 2015, up to 70% of motorized travel in Dakar was on public transport. The number of daily passengers on TER Phase 1 increased by 220% over that of the PTB.

Transport safety

The evaluation found that air transport projects targeting safety contributed to improving

aviation safety records in the beneficiary countries and the Central Africa region.

In the case of the Priority Air Safety Project in the Democratic Republic of Congo, the annual average number of air transport incidents/accidents declined from 10 in 2013 to 3 in 2022. This is attributable to the increase in the rate of airspace coverage by adequate surveillance and air navigation equipment from 95% in 2013 to 100% in 2022. Similarly, at the regional level, the proportion of major international airports with runways offering optimal service levels and adequate approach equipment rose from 60% in 2013 to 100% in 2022. Despite this progress, regional aviation safety oversight agencies continue to face challenges in building their capacity, primarily due to dependence on external financing from regional economic communities.

Capacity and institution building

The evaluation of the cluster projects demonstrated their successful contribution to developing and enhancing the capacities of institutions in both the rail and aviation sub-sectors. For example, the Creation of a Leasing Platform for African Airlines Project supported the development of a strategic plan for the

operationalization of the proposed leasing platform. Regarding this, a technical workshop was convened in mid-2022 and met or exceeded its engagement targets according to the interviewees. This workshop brought together several stakeholders, including financial institutions, original equipment manufacturers (OEMs), airlines, airline associations and other relevant independent participants, thus encouraging the collation of diverse views from those most likely to engage with the leasing platform once launched. Recognizing the value added by the broader industry's participation, further engagements beyond the original scope of the project were held with additional OEMs, banks, and an aircraft lessor at a continental aviation summit. Progress has been made in developing the leasing platform's operationalization strategy.

Efficiency

Efficiency measures the quality of project implementation relative to timeliness, resource use (adherence to budgets and efficient utilization of funds), and soundness of the project's economic rationale (cost-benefit analysis).

Figure 6: Summative project ratings – Efficiency

Corporate Loan to Ethiopian Airlines	
Nacala Rail Corridor and Port Project & Value Addition Project	N/A
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	
Air Côte d'Ivoire Modernization and Expansion Program	
Ghana Airports Company Limited (GACL) Capital Investment Programme	
Morocco - Railway Infrastructure Reinforcement Project	
South Africa - Corporate Loan to Transnet	
Tunisia - Rail Infrastructure Modernization (II)	
Democratic Republic of Congo - Priority Air Safety Project Phase II	
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	
Support for the Air Transport Sectors of West & Central Africa	
Support to AUC/AFCAC on Single African Air Transport Market	
Creation of a Leasing Platform for African Airlines	
Sharm El-Sheikh Airport Development Project	N/A
Jomo Kenyatta International Airport Airfield Expansion	N/A
Feasibility Study on the Extension of the Cameroon-Chad Railway	

Overall, the evaluation assessed the efficiency of the cluster projects as partly unsatisfactory.

In terms of timeliness, the evaluation found the majority (11/13) of the implemented projects to have recorded significant delays, leading to cost overruns in some cases, particularly the Priority Air Safety Project Phase II in the DRC. In terms of adherence to budgets, the cluster projects largely remained within their approved budgets, albeit with a slight cost overrun which, however, was mostly not significant enough to cause the non-delivery of the outputs. The efficient utilization of funds could not be fully assessed due to the unavailability or limited access to financial records of some of the beneficiary organizations.

The non-sovereign operations could not be rated on project efficiency using the same judgment criteria, as the clients did not disclose data on investment costs.

Individual project ratings for efficiency are presented in Figure 6.

Timeliness

Project delays were a common challenge across both sub-sectors based on document reviews

(PCRs and XSRs), but particularly for rail. Rail projects experienced an average delay of 17.4 months between approval and disbursement, while aviation projects had shorter delays at 8.1 months. Delays in implementation impacted project timelines and disbursement rates, and undermined the timely delivery of critical infrastructure, hindering the sub-sectors' full economic potential. See Table 6.

The evaluation rated the timeliness of the project implementation across both sub-sectors as unsatisfactory (Table 7). Of the 13 projects that had either been delivered or were being implemented, only two have not been affected by delays during implementation (the Ghana Airports Company Limited (GACL) Capital Investment Programme, and the Corporate Loan to Transnet). **The remaining projects all recorded delays of between 17 and 60 months beyond their scheduled implementation periods.** The project delays, measured as the gap between the originally planned completion date and the actual (or expected) completion date, are shown in Table 7.

Interviewees noted that the delays stemmed from several sources, including the administrative decision-making processes defined by the delegation

Table 6: Average delay in months from funding approval to further implementation of transport projects' operations by sub-sector (2012-2023)⁹

Sub-sector	Delays between approval and project entry into force		Delays between signature and project entry into force		Delays between planned first disbursement and actual first disbursement		Delays between approval date and planned first disbursement		Delays between planned final disbursement and latest disbursement	
	No. of project operations	Months (average)	No. of project operations	Months (average)	No. of project operations	Months (average)	No. of project operations	Months (average)	No. of project operations	Months (average)
Air Transport	25	4.5	23	1.7	22	9.0	22	8.1	22	19.4
Rail Transport	11	9.2	11	0.9	10	4.5	11	17.4	11	11.4

Source: IDEV calculation, based on PCRs/Project Performance Evaluation Reports available for comparison.

Table 7: Assessment of projects' implementation timeliness

Project Name	Original planned completion date	Actual/ expected completion date	Delay
Support for the Air Transport Sectors of West & Central Africa	06/2019	06/2024	60 months
Support to AUC/AFCAC on Single African Air Transport Market	03/2023	12/2024	21 months
Creation of a Leasing Platform for African Airlines	09/2022	12/2024	27 months
Feasibility Study on the Extension of the Cameroon-Chad Railway	04/2020	07/2024	51 months
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	03/2022	12/2024	33 months
Air Côte d'Ivoire Modernization and Expansion Program	07/2022	12/2023	17 months
Ghana Airports Company Limited (GACL) Capital Investment Programme	06/2018	06/2018	None
Corporate Loan to Ethiopian Airlines	01/2029	01/2024	60 months
Morocco - Railway Infrastructure Reinforcement Project	05/2019	12/2022	43 months
Democratic Republic of Congo - Priority Air Safety Project Phase II	12/2014	06/2019	54 months
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	07/2019	12/2021	29 months
South Africa - Corporate Loan to Transnet	07/2036	N/A	None
Tunisia - Rail Infrastructure Modernization Project (II)	12/2008	08/2012	44 months

Source: Project assessments (data from project completion reports and site visits).

of authority matrix of the Bank, which grants specific authority to local staff, the need for non-objection notices for all contractual changes, and capacity limitations of executing agencies. Other causes of delay included inefficient coordination between implementation teams, slow release of counterpart funding, complicated and inflexible Bank procedures relating to procurement, slow disbursement, complexity and significance of projects, and the impacts of the COVID-19 pandemic.

Project delays were not unique to the cluster projects, nor the rail and aviation sub-sectors, with the broader transport sector portfolio analysis highlighting the frequency of delays experienced across all of the Bank's transport projects (Box 4).

The delivery of projects was complicated further by challenging circumstances beyond the control of the Bank and the beneficiaries — most notably the onset of the COVID-19 pandemic and terrorism. For instance, the Sharm El-Sheikh Airport Development Project was ultimately cancelled due to a significant and sudden decline in demand for flights to and from the airport caused by the 2005 major terrorism incident (Sharm El-Sheikh bombings). The Jomo Kenyatta International Airport

Airfield Expansion Project was cancelled in response to consequential changes in the competitive landscape of East Africa's regional aviation market and the view that proceeding with the project would have heightened Kenya Airways' exposure to further competition from major Middle Eastern airlines.

Resource use

The evaluation rated resource use across both sub-sectors as satisfactory. The evaluation found that despite implementation delays, more than half (8/13) of the implemented cluster projects obtained high ratings in terms of judicious use of financial resources. It is worth noting that the information available in the project documentation provided did not allow for conclusions to be made around financial/budgetary performance in five of the desk-based assessments. Also, the cancellation of three projects prior to implementation precluded analysis relating to their budgetary performance. Of the remaining projects, most were viewed favourably in their use of resources — receiving scores reflective of either satisfactory or highly satisfactory results. Given that most projects experienced delays, the ability to deliver results without significant cost overruns is commendable.

Box 4: Transport project implementation delays in the portfolio analysis (2012–2023)

As per the table below, the Bank's broader portfolio of transport projects was exposed to frequent delays across all sub-sectors, with projects averaging 1.3 years beyond their planned completion dates. Over 60% of all projects had their closing dates extended more than once. With 1.67 average project extensions per railway project, on-time delivery seemed to be a particular issue within the sub-sector. These delays originated at different stages of the projects and had several causes, including operational deficiencies, long procurement times and inertia within and between organizations, and were compounded by the effects of the COVID-19 pandemic. The delays culminated in almost half of all projects receiving either partly unsatisfactory or unsatisfactory ratings for timeliness.

	Aviation	Rail	Ports / Water	Roads	Urban transport	Multi-sector
No. of projects	20	11	19	166	9	13
No. of extensions	26	20	27	409	17	21
No. of projects w/o extension	9	3	7	69	1	4

Source: Calculated by IDEV, based on Bank internal databases (as of November 2024).

Despite most of the cluster projects attaining satisfactory ratings for resource use efficiency, eight (8/13) implemented cluster projects received partly unsatisfactory scores relative to the sub-criteria. In the case of the Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I, cost overruns exceeded 60%. Although the core project components (i.e., those relating to the development of the railway system itself) were achieved, ancillary works had not been fully delivered, despite these significant cost overruns. Similarly, in the case of the Democratic Republic of Congo - Priority Air Safety Project Phase II, cost overruns were incurred due to procedural breaches by the executing agency, notably the absence of a notice of non-objection on project amendments. Given that the need for non-objection notices from the Bank has been a source of several delays in the projects assessed, the ineffectiveness of the Bank's safeguarding processes in preventing this breach and its associated cost implications is noteworthy.

The evaluation's assessments of IPRs found shortcomings relative to cost overruns and financial management challenges. These were found to be linked to changes in project designs, limited guidance and supervision of executing agencies, and their ability to comply with the Bank's procedural requirements,

whose implementation was considered by some executing agencies as less flexible and not commensurate with the scale, complexity, and significance of undertakings. For example, in the case of the Institutional Support to the African Civil Aviation Commission for the Implementation of the Single African Air Transport Market, disbursement of resources was significantly slower than initially intended. According to the latest (December 2023) IPR, approximately half of all disbursements were rejected by the Bank, implying a lack of compliance with its requirements. Financial planning in projects was further complicated by uncertainty and/or a lack of detailed project planning at the time of approval. Unforeseen technical challenges relating to delivery often necessitated amendments to preliminary project plans, with associated budgetary implications. Additionally, the lengthy lead-times for obtaining non-objection notices of budgetary amendments, where needed, also created a perverse incentive for financial contingencies to be inflated in project preparatory phases.

Soundness of the project's economic rationale

The evaluation had a limitation in undertaking a thorough reassessment of the project economic rate of return (ERR) for the cluster projects, even

though preliminary projections at appraisal pointed to the likelihood of economic gains for most projects. Some projects were either ongoing, cancelled, or lacked details in the documents reviewed to be used to calculate the economic rate of return. Even though the evaluation could not undertake a rigorous assessment of the ERR to be used to calculate project net benefits, the projects were found to be based on sound economic objectives.

Sustainability of Results

The assessment of sustainability considers the extent to which the projects have addressed risks during implementation and put in place mechanisms to ensure the continued flow of benefits after completion. Such an analysis is based on five separate elements: (i) technical soundness; (ii) financial sustainability; (iii) institutional sustainability and strengthening of capacities; (iv) ownership and sustainability of partnerships; and (v) environmental and social sustainability (only applies to Environmental and Social Category I and II projects).

On balance, the evaluation rated the sustainability of the cluster projects as satisfactory, while noting strong variability across projects. The evaluation found that eight out of 14 (8/14) projects were likely to ensure that their benefits would be resilient over the medium and long terms. However, six projects were rated partly unsatisfactory or unsatisfactory due to shortcomings in several measures, including long-term financial sustainability, weak institutional capacity, and weak environmental and social safeguards. The assessment considered the factors contributing to sustainability, including strong sponsors that backed up some of the projects, the financial power of the sponsors, as well as the design and delivery of high-quality outputs. At an aggregate level, the evaluation results for sustainability varied across projects, as seen in Figure 7.

Technical soundness

From a technical perspective, most physical infrastructure designs were found to reflect

Figure 7: Summative project ratings – Sustainability

Corporate Loan to Ethiopian Airlines	
Air Côte d'Ivoire Modernization and Expansion Program	
Morocco Railway Infrastructure Reinforcement Project	
Ghana Airports Company Limited (GACL) Capital Investment Programme	
Creation of a Leasing Platform for African Airlines	
Dakar-Diamniadio-AIBD Regional Express Train Project - Phase I	
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	
South Africa Corporate Loan to Transnet	
Tunisia - Rail Infrastructure Modernization Project - Phase II	
Nacala Rail Corridor and Port Project & Value Addition Project	
Feasibility Study on the Extension of the Cameroon-Chad Railway Project	
Support to AU C/AFCAC on Single African Air Transport Market	
Sharm DEI-Sheikh Airport development Project	N/A
Jomo Kenyatta International Airport Airfield Expansion	N/A
Support for the Air Transport Sectors of West & Central Africa	
Democratic Republic of Congo - Priority Air Safety Project Phase II	

appropriate quality standards and included appropriate adaptation elements. Projects involving the development of physical infrastructure appeared to be designed with appropriate quality standards in mind. The projects typically also included consideration for the long-term effects of climate change, such as heightened temperature fluctuations and increases in the prevalence and intensity of flooding, in technical designs. This was reflected in the evaluation scoring, with all relevant projects being viewed as either satisfactory or highly satisfactory with respect to their technical soundness.

Financial sustainability

The assessment of the cluster projects on this sub-criterion revealed insufficient funding and resource mobilization for operations and maintenance of projects' facilities for some of the projects. In many cases, the resources available or anticipated are not sufficient to cover maintenance costs, which may be a threat to sustainability. Long-term financial sustainability concerns have the potential to undermine efforts to deploy appropriate and adequate maintenance programmes that extend beyond the Bank's involvement, thus threatening the infrastructure's ability to deliver benefits over its full design lifespans. Lack of appropriate financial planning and resource mobilization for operation and maintenance hence represent a serious threat to projects' technical sustainability. For example, financial sustainability with respect to the Project for Support to the Air Transport Sectors of West and Central Africa Project was largely dependent on the effectiveness of the new agencies' funding mechanisms, which could only be guaranteed if they had sufficient funding from airport fees and charges, to generate sustained investments and continuously improve airport services. The operationalization of these new agencies relied on the fulfilment of capacity-building objectives set out in this project, which involved training of senior staff and regional inspectors for such structures, and organizational support in information technology (IT) and management.

Long-term continuity established by clearly defined contractual arrangements could offer stability both from a financial and institutional standpoint. Several assessments raised concerns about the long-term financial sustainability, with potential consequences for infrastructure maintenance. These concerns stem from limitations in projects' revenue-generating potential and the associated reliance on subsidization for cost recovery, as well as deficient financial management among some executing agencies. In the case of the Nacala Rail Corridor and Port Value Addition Project, clear contractual arrangements relating to ownership, operation and maintenance encouraged relatively predictable revenue generation and benefit realization during the time horizon covered by the concession agreements.

Institutional sustainability and capacities

The project assessments found variations in institutional capacity and strength among executing agencies and other relevant participants. While some beneficiary institutions have adequate implementation capacity; others have low capacities mainly due to lack of or limited project implementation experience. For example, in the case of the Morocco Railway Infrastructure Reinforcement Project, the executing agency was praised for its internal institutional capacity and track record of stability. In contrast, the assessment of the Jomo Kenyatta International Airport Airfield Expansion Project noted that institutional challenges faced by the Government of Kenya and the Kenya Airports Authority posed risks to the project's development and long-term operation. This suggests the need for institutional reviews to form part of risk assessments and mitigation strategies, which would assist the Bank to gauge whether close supervision and/or capacity-building initiatives would complement its investments in physical infrastructure on a case-by-case basis. Similarly, a lack of continuity was noted in some capacity-building initiatives. For example, in the assessment of the Nacala Rail Corridor and Port Value Addition Project, it was highlighted that the entities responsible for long-term capacity building

did not necessarily have sufficient resources to continue their role beyond the conclusion of the project.

Ownership and sustainability of partnerships

The assessments of individual cluster projects highlighted strong ownership by the beneficiary institutions and governments. They noted cases of well-established institutions with clear mandates, effective rail and aviation maintenance policies, planning, and organization, including autonomous maintenance funds, and with beneficiaries having demonstrated a strong commitment to sharing project costs, including operations and maintenance. Some of the beneficiaries have adopted strategies, including the designation of operations and maintenance responsibilities. For example, in the Project for Support to the Air Transport Sectors of West and Central Africa (PASTA-CO), the coordination between regional agencies in the African aviation industry, prior to the project's implementation, to ensure alignment with ECOWAS, ECCAS and WAEMU-related initiatives was illustrated by the assistive efforts they exhibited to prepare ICAO audits and certifications, and to supervise airports and airlines. The coordination efforts between the regional economic communities (RECs) and ICAO appear to have been sufficient, though often undocumented. Similarly, from the onset of the Dakar-Diamniadio-AIBD Regional Express Train Project – Phase I, the Government of Senegal demonstrated strong ownership and commitment to the project. In pursuance of this, the implementation was managed by the Senegalese government's investment promotion agency, Promotion des Investissements et Grands Travaux (APIX). After the construction, the management of the TER was entrusted to Société Nationale de Gestion du Patrimoine du Train Express Régional (SENTER) to ensure continued ownership. Long-term sustainability seems to be generally secured, especially with the Phase 2 operating cost contribution covered by government reducing to 10%-20%, with annual allocation towards sustaining capital remaining sufficient. The strong partnership established under the project yielded a co-financing

arrangement with Islamic Development Bank and the French Government, culminating in meeting the capital cost overrun of Phase 1 when the initial budget was exceeded by 62.3%.

Environmental and social sustainability

The assessments highlighted the importance of environmental sustainability of aviation projects. Multiple in-depth project analyses found that adequate measures have been considered in the designs of the projects to mitigate any adverse environmental effects on the beneficiary populations. However, the implementation of the mitigation measures to offset the increases in GHG emissions was described as limited and not necessarily deliberate. Therefore, while greenhouse gas emissions could be expected to increase because of the Bank's projects, this should be viewed in the context of the significant socio-economic benefits afforded by the Bank's investments in improving the connectivity, safety and competitive landscape of Africa's aviation sub-sector as well as the context of the Paris Agreement, whose country-specific targets allow for slight increases in emissions.

Despite efforts to improve environmental and social sustainability, several cluster project assessments found insufficient evidence on these aspects to confirm that such sustainability was secured in the medium and long term. For example, in the Nacala Rail Corridor and Port Project and Value Addition Project, employment targets were generally met, but metrics related to women employed were not provided, and while modal shifts from road to rail were reported, evidence was anecdotal in nature. In the Tunisia Rail Infrastructure Modernization Project, it was also noted that little evidence was available to demonstrate improvement in healthcare, education, women and youth employment.¹⁰

Additionality of Non-Sovereign Operations

The additionality component of the evaluation considers whether non-sovereign projects could have been delivered without support from the Bank

with comparable quality, scope and timeframes. Of the 16 projects evaluated, five were non-sovereign and hence reviewed in terms of their additionality.

The Bank's presence in non-sovereign operations was found to have considerable added value. Due to the Bank's credit ratings and track record, its involvement in projects allayed concerns around political risk, particularly in frontier markets that might otherwise not have attracted commercial capital. The additionality that the Bank can offer was also considered highly favorable by its clients and encouraged buy-in from commercial clients. The project assessments highlighted that the Bank's financial involvement in projects mitigated concerns held by commercial financiers around the political risks associated with the projects' locales. This ability to ease perceptions of risk — underpinned by the Bank's favourable credit ratings¹¹ and the depth of its experience in financing and overseeing large-scale capital-intensive projects on the continent — allowed it to coordinate and structure resources that might otherwise not have been availed in frontier markets. This is illustrated in Box 5, which presents an extract from the assessment of the Nacala Rail Corridor and Port Project & Value Addition Project.

In addition, the challenge in attracting potential financing partners for capital-intensive projects was addressed by the involvement of the Bank, with its financing terms and magnitude of funding contributing to the commercial viability of projects and their beneficiaries' operations. In the case of the Ghana Airports Company Limited Capital Investment Programme Project, although the executing agency had received funding from the private sector prior to the Bank's involvement, both the value of the Bank's commitment and its terms were considered adequate in meeting the funding needs of the client. In addition, the Bank's reputation as both a trusted financier and facilitator of collaboration in the Creation of a Leasing Platform for African Airlines project placed it in a strong position to mobilize and coordinate resources — whether financial or otherwise. Without support

from the Bank, individual airlines — particularly the relatively small airlines that the leasing platform intends to support — may lack the capacity to coordinate efficiently among each other without external support. Exacerbated by the financial consequences of the COVID-19 pandemic, small airlines may also have lacked access to the funding needed to undertake studies and develop strategies akin to those that form part of this project.

Furthermore, the Project Performance Assessment Report (PPAR) of the AfDB's Corporate Loan to Ethiopian Airlines stated that the Bank provided the loan at a time when the UK Export Credit Agency (ECA) financing was not available. The Bank provided Ethiopian Airlines a timely, cost-effective alternative, which helped avoid delays in the implementation of its expansion and modernization plan. The Bank proactively identified and resolved problems at different stages of the project cycle, according to the project performance assessment report. At the time of arranging the loan, the UK ECA had suspended financing, so the Bank proposed a Corporate Loan with credit insurance issued by the African Trade Insurance Agency, which was 95% indemnified. This enabled the loan agreement and the project to proceed.

The PARs of the cluster projects showed that without adequate mitigation measures or reparation, investments in physical transportation infrastructure often had the potential to severely degrade the local environment and disrupt the ways of life of residents in affected areas. Through the Bank's negotiating leverage when defining borrowing terms, it could insist that borrowers comply with its Environmental and Social Operational Safeguards, as set out in its Integrated Safeguards System. Relatedly, as alluded to above, the transport projects that the Bank supports often create opportunities for complementary components and separate value-addition projects that seek to promote environmental and social causes in the affected areas. In doing so, the Bank adds value not only as a financier, but

Box 5: The Bank's added value in risk perception

Owing to the magnitude of capital required for the Nacala Rail Corridor and Port Project and Value Addition Project as well as perceptions of political risks, it is unlikely that commercial lenders would have financed the project on terms feasible for the borrowers. Unlike commercial lenders, the Bank could arrange financing over an extended time horizon at a competitive price point. In addition to providing USD 300 million in funding, the Bank played a central role alongside the International Finance Corporation and Japan Bank for International Cooperation in the financial structuring of the project. In this role, the Bank led negotiations with governments and ensured that the project would comply with its Integrated Safeguards System. Because of the Bank's reputation, its investment in the project also allayed concerns around political risk held by other development partners, thereby encouraging buy-in.

Source: Assessment of P-Z1-D00-032/033/034/035 – Multinational – Nacala Rail and Port Project & P-MW-KB0-002 - Malawi – Nacala Rail and Port Value Addition Project.

as a promoter of sustainable development. This is illustrated in Box 5, which presents an extract from the evaluation of the Nacala Rail Corridor and Port Project & and the Port Value Addition Project.

Assessment of Cross-cutting Issues

This section presents an in-depth assessment of how the cluster projects integrated important cross-cutting issues, including gender, climate, and state fragility, in their design and delivery frameworks.

Gender

The evaluation noted shortcomings regarding investments in physical infrastructure whose designs considered the specific needs of women, youth, the elderly, and people with disabilities.

The project assessments found that physical infrastructure was generally developed to suit the needs of the broader market, while gender-specific benefits were to be delivered principally through ancillary components that sought to leverage the improved connectivity or job opportunities afforded by the infrastructure. While acknowledging that there may have been opportunities for the Bank to further enhance its gender mainstreaming efforts through adopting gender-sensitive designs, the evaluation generally found that a gender lens was applied at the conceptual level of project planning through the inclusion of ancillary components intended to deliver benefits to women, youth, elderly and people living with disability specifically. These components

included, among others, the construction of women's centers, provision of training opportunities for women and support for women's groups.

However, although the cluster projects typically included gender-related ancillary components in their plans, these components were not always fully realized and, in some instances, were cancelled altogether, suggesting that the Bank sometimes fell short of its stated objectives. Relatedly, efforts to ensure long-term continuity of these ancillary components were limited. The cluster evaluation also noted that indicators to measure the intended impacts of projects on women, youth and the vulnerable were either not always included in logical frameworks or reported on in the documentation made available, limiting the extent to which the success of gender-related initiatives could be assessed.

Climate

The designs of the cluster projects involving physical infrastructure were found to have included adequate consideration for the growing risks posed by climate change, such as increases in the intensity and frequency of extreme weather events, thereby raising confidence in the resilience of the infrastructure. Despite the environmental and climate adaptation and mitigation measures incorporated into the designs of projects, monitoring of environmental impacts was noted as a challenge in more than one project. In others, particularly from the aviation sub-sector,

it was noted that although monitoring either took place or would have taken place (i.e., in the case of cancelled projects), there was limited evidence to suggest that this monitoring translated to any form of modifications. Most projects from the aviation sub-sector did not appear to have strategies aimed at offsetting the net increases in GHG emissions that the interventions would bring about. In the majority of the cluster projects, the net increases in GHG emissions were considered by the beneficiaries alongside the significant economic and connectivity benefits afforded by regional aviation infrastructure — particularly in the context of Africa’s emerging aviation market. A typical case is the corporate loan to Ethiopian Airlines.

State fragility

Of the five projects in states facing fragility, two were found to have had clear benefits for such states. These benefits were usually related to poverty alleviation through the creation of direct and indirect employment opportunities, improved connectivity in remote locations, enhanced regional connectivity, and improved integration into international markets. While other projects were not necessarily expected to deliver direct benefits that would aid in mitigating state fragility, their potential to deliver indirect benefits was deemed clear. Table 8 presents a list of projects implemented in states facing fragility. ■

Table 8: Projects undertaken in states facing fragility

Project Name	Description of fragility mitigation
Nacala Rail Corridor and Port Project & Value Addition Project	- Enhanced access to economic opportunity and food security through the value addition project. - Enhanced connectivity in remote regions and between countries along the development corridor through core project components.
Democratic Republic of Congo - Priority Air Safety Project Phase II	- Strengthening of regional aviation infrastructure. - Enhancement of safety among airlines. - Efforts to remove airlines from EU Blacklist, with resultant impacts on international integration.
Feasibility Study on the Extension of the Cameroon-Chad Railway	No direct impacts on fragility as these were merely studies.
Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway	Potential to improve access to economic opportunity and international markets among multiple fragile states, with resultant in-flows of foreign exchange if railways were to be operationalized.
Corporate Loan to Ethiopian Airlines	- No direct impacts on fragility. - Ethiopian Airlines is a source of significant foreign exchange, and the expansion of the airline enhances Addis Ababa’s position as a major aviation hub in Africa, with clear benefits for employment and international integration that extend beyond Ethiopia’s borders.

Lessons

The following lessons were identified by the evaluation:

Lesson 1: Enhancing Project Performance

The Bank's technical support and close supervision of projects involving partner executing agencies with limited implementation capacity can significantly enhance project performance. Project assessments revealed varying levels of institutional capacity and strength among executing agencies. For example, in the case of the Morocco Railway Infrastructure Reinforcement Project and the Sharm El-Sheikh Airport Development Project, the executing agencies were praised for their internal institutional capacity and track record of stability. In contrast, the assessment of the Jomo Kenyatta International Airport Airfield Expansion Project noted that institutional challenges faced by the Kenya Airports Authority posed risks to the project's development and long-term operation. Implementation capacity assessment at entry is therefore important to ascertain to what extent the executing agency has the requisite implementation capacity. Assessing and addressing the institutional capacity of executing agencies at the outset of projects is key to enhancing project performance. Where capacity is found to be deficient, including Bank technical support and close supervision as a component of the project interventions, can help to address the challenges and enhance project performance.

Lesson 2: Demonstrating Additionality

Given the capital-intensive nature of investments in the railway and aviation sub-sectors, effectively demonstrating Bank additionality is paramount. This ensures that the limited resources of the Bank are allocated appropriately and allows the Bank to highlight its unique contributions beyond what is available in the market, ensuring it does not crowd

out the private sector and that it complements other financiers. For example, the Bank can play a crucial role by offering long-term financing for high-risk, large-scale infrastructure projects that commercial banks typically avoid. Furthermore, the AfDB can provide essential non-financial support, such as technical expertise, policy advice, and capacity building, to enhance project design and implementation.

Lesson 3: Financial Sustainability

A well-thought-out long-term financial sustainability plan for capital-intensive operations such as railways and aviation projects can enhance efforts to implement appropriate and adequate maintenance programs that extend beyond the Bank's involvement.

Lesson 4: Robust M&E Systems

A robustly designed and implemented monitoring and evaluation (M&E) system is crucial for the successful delivery of interventions, as it helps track progress and identify areas needing improvement. However, project assessments found that while logical frameworks were generally well-articulated, the quality and adequacy of M&E systems and their implementation were often insufficient. Additionally, limitations were identified in project risk identification and mitigation, as well as in monitoring implementation progress. Issues related to the inadequacy and choice of outcome indicators and their monitoring were highlighted in multiple assessments. Despite including results-based management arrangements at the design stage, the design and use of M&E were found to be inadequate. For instance, some project implementation arrangements lacked well-defined logic models or frameworks for assessing outcomes and impacts. In some cases, there were no monitoring plans with measurable indicators and baselines at appraisal,

limiting the evaluability of projects at midline and endline. For example, in the Support to AUC/AFCAC on Single African Air Transport Market Project, the links between the intervention and the selected impact indicators were noted as tenuous and exposed to external influence, undermining efforts to monitor both implementation progress and the projects' contribution towards achieving their core purpose. Ensuring that the M&E system avoids these various shortcomings can promote timely data analysis for accountability, management, and learning.

Lesson 5: Environmental and Social Safeguards

Implementing adequate environmental, social, and operational safeguards for investments in physical infrastructure can prevent potential environmental degradation and protect the ways of life of beneficiary communities. The evaluated cluster projects were found to have largely created

opportunities for complementary components and separate value-addition projects that promote environmental and social causes in the affected areas. By doing so, the Bank added value not only as a financier but also as a promoter of sustainable development. For example, in addition to providing USD 300 million in funding for the Nacala Rail Corridor and Port Project & and Port Value Addition Project, the Bank played a central role alongside the International Finance Corporation and Japan Bank for International Cooperation in the financial structuring of the project, leading negotiations with governments and ensuring compliance with its Integrated Safeguards System. The Bank's reputation helped allay concerns around environmental, social, and political risks held by other development partners, encouraging their buy-in. However, compliance with the safeguards by the borrower may be unlikely if this obligation is not defined in the borrowing terms. ■



ATF2024

AFDB TRANSPORT FORUM
FORUM TRANSPORT DE LA BAD



AFRICA ON THE MOVE – ACCELERATING
SUSTAINABLE TRANSPORT
AND LOGISTICS CONNECTIVITY

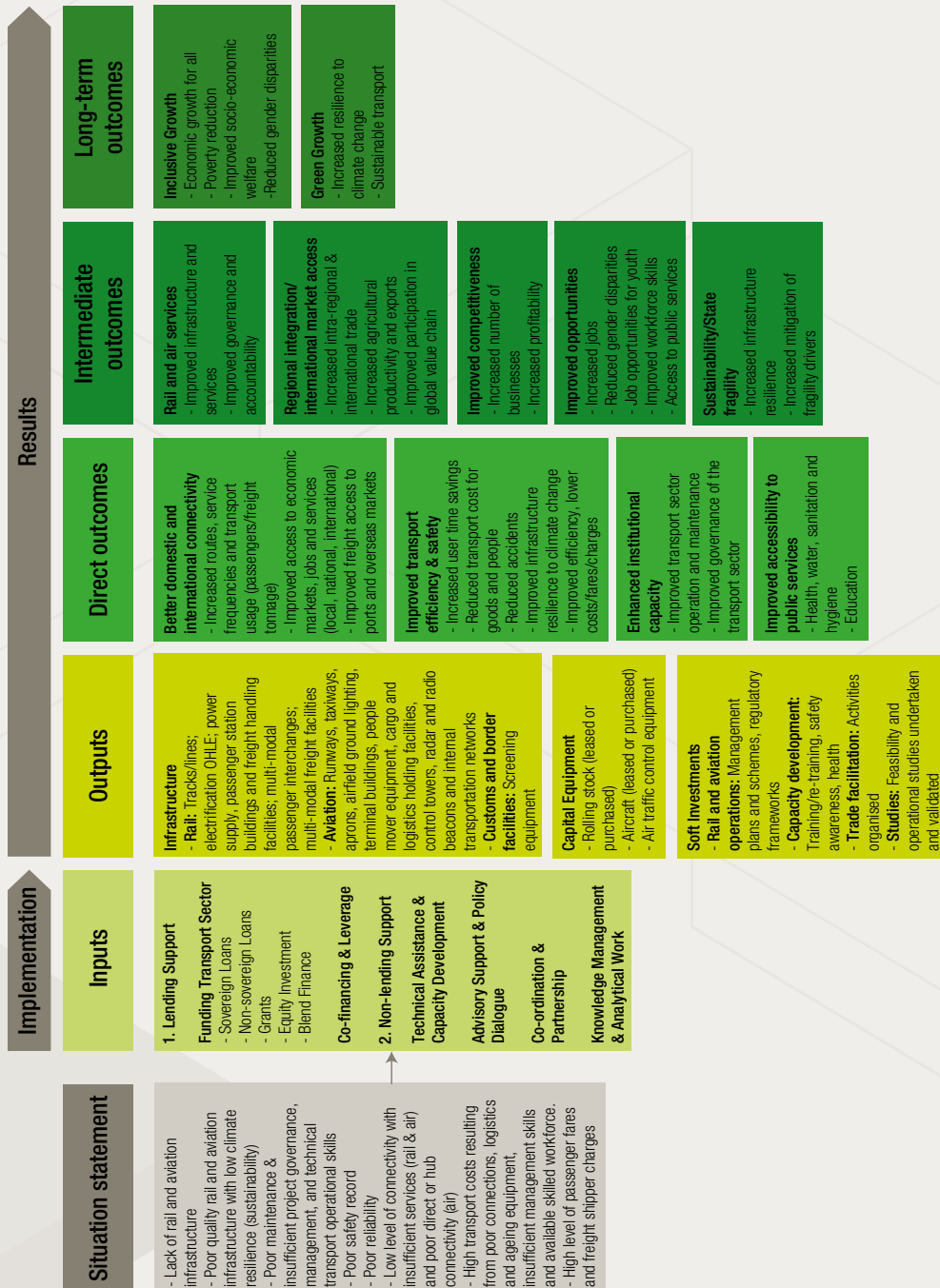
L'AFRIQUE EN MOUVEMENT – ACCELERER
LA CONNECTIVITE
ET DE LA LOGISTIQUE



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Annex 1: Rail and Aviation Cluster Evaluation Theory of Change



Annex 2: Rating Grid for Project Assessment

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
Relevance			
Relevance of Objectives			
Were the project objectives, as outlined in the Project PAR, aligned and consistently maintained with the overarching objectives of the African Development Bank (AfDB) as reflected in the Ten-Year Strategy (TYS) and Country Strategy Paper (CSP)?	JC1.1 The project objectives (per the Project PAR) were and remained in line with the AfDB objectives (TYS, CSP).	1 = During the implementation period, the project purpose was not aligned with either: i) the Bank's TYS ii) the Country Strategy Paper. 2 = During the implementation period, the project purpose was only somewhat aligned with one of: i) the Bank's TYS ii) the Country Strategy Paper. 3 = During the implementation period, the project purpose was largely aligned with one of: i) the Bank's TYS ii) the Country Strategy Paper. 4 = During the implementation period, the project purpose was fully aligned with both of the following: i) the Bank's TYS ii) the Country Strategy Paper.	
Did the project objectives, as detailed in the Project PAR, consistently align with the High 5s, African Union Agenda 2063, Sustainable Development Goals (SDGs), and relevant sector strategies throughout the project lifecycle?	JC1.2 The project objectives (per the Project PAR) were and remained in line with the High 5s, African Union Agenda 63, SDGs and sector strategies.	1 = During the implementation period, the project purpose was not aligned with either: i) High 5s ii) African Union Agenda 63 iii) SDGs and iv) sector strategies. 2 = During the implementation period, the project purpose was somewhat aligned with one or more of: i) High 5s ii) African Union Agenda 63 iii) SDGs and iv) sector strategies. 3 = During the implementation period, the project purpose was largely aligned with at least three of: i) High 5s ii) African Union Agenda 63 iii) SDGs and iv) sector strategies. 4 = During the implementation period, the project purpose was fully aligned with all of the following: i) High 5s ii) African Union Agenda 63 iii) SDGs and iv) sector strategies.	
To what extent has the project been responsive to the evolving needs of RMCs, RECs and the local populations (including women)?	JC1.3 The project objectives responded to country needs including local population, including women and young people.	1 = During the implementation period, the project was barely aligned with any of the following i) country's development strategies, ii) transport investment plans/ investment gaps, iii) local needs. 2 = During the implementation period, the project had some alignment with one of the following i) country's development strategies, ii) transport investment plans/ investment gaps, iii) local needs. 3 = During the implementation period, the project was largely aligned with one of the following i) country's development strategies, ii) transport investment plans/ investment gaps, iii) local needs. 4 = During the implementation period, the project was fully aligned with one of the following i) country's development strategies, ii) transport investment plans/investment gaps, iii) local needs.	

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
To what extent could the project have been implemented without the AfDB input (financial and nonfinancial) with the same scope, quality and/or timeframe?	JC1.4 AfDB input was required to create project scope, quality, and within allotted timeframe; implementation without AfDB input would be uncertain.	1 = The project would have continued as it did without AfDB (potentially funded by another organization). 2 = The project would have continued at the same scale and speed, but not been aligned to key AfDB objectives. 3 = The project would have continued but at smaller scale or slower. 4 = No transport infrastructure project would have been delivered without AfDB involvement.	
Score for Relevance 1			
Relevance			
Relevance of Design			
Considering predictable factors, did the project design ensure effective outputs and outcomes that were in line with the stated objectives, discounting unpredictable external stimuli?	JC1.5 Project designs are appropriate (discounting unpredictable external stimuli), ensuring effective outputs/outcomes and alignment with objectives.	1 = Project design activities were limited and only had one of the following: i) strong rationale (showing causal chains) ii) risk mitigation strategy iii) effective monitoring system. 2 = Some project design activities were conducted and had two of the following in some order: i) strong rationale (showing causal chains) ii) risk mitigation strategy iii) effective monitoring system. 3 = The project design activities were considered and had two of the following in good order: i) strong rationale (showing causal chains) ii) risk mitigation strategy iii) effective monitoring system. 4 = The project design activities were fully thought through and had all of the following: i) strong rationale (showing causal chains) ii) risk mitigation strategy iii) effective monitoring system.	
In cases where applicable, to what extent did the project design integrate the unique characteristics and challenges of fragile states?	JC1.6 The project design integrates the peculiar characteristics of fragile states (only where relevant).	1= The intervention design encompasses/considers just one or none of the above factors. 2= The project only includes 2 or 1 of the following elements: i) the project was designed applying a fragility lens; ii) the project contributes to reduce fragility; iii) the project includes components/activities that address fragility iv) the project has specific indicators for fragility issues. 3= The project only includes 3 of the following elements: i) the project was designed applying a fragility lens; ii) the project contributes to reduce fragility; iii) the project includes components/activities that address fragility iv) the project has specific indicators for fragility issues. 4= i) The project was designed applying a fragility lens; ii) the project contributes to reduce fragility; iii) the project includes components/activities that address fragility iv) the project has specific indicators for fragility issues.	

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
Regarding the intervention's design, assess the clarity and focus of the objectives. Evaluate the realism of intended outcomes given the country's current circumstances, the quality of risk assessment, appropriateness of solutions, relevance of modifications, and consideration of prevailing circumstances.	JC1.7 The intervention's designs has clear objectives: (i) The extent to which the intervention's objectives are clearly stated and focused on outcomes as opposed to outputs; (ii) The realism of intended outcomes in the country's current circumstances; (iii) The quality of risk assessment (assumptions made in the logic model); (iv) The extent to which intervention design adopted the appropriate solutions to the identified problems; (v) The relevance of modifications made to intervention design; (vi) The circumstances prevailing at the time of the evaluation. The evaluator should assess to what extent potential negative impacts were identified, their likelihood of occurring and how they might be avoided.	1= The intervention design encompasses/considers just one or none of the above factors. 2= The intervention design encompasses/considers a few of the above factors. 3= The intervention design encompasses/considers most of the above factors. 4= The intervention design encompasses/considers all of the above factors (i) to (vi).	
How did the project design adapt over time in response to changing and unpredictable external stimuli?	JC1.8 The design of the project altered over time in response to changing and unpredictable external stimuli.	1 = The project design did not respond to unpredictable external stimuli. 2 = The project design altered slowly to somewhat respond to unpredictable external stimuli. 3 = The project design altered quickly to somewhat respond to unpredictable external stimuli. 4 = The project design altered quickly to fully respond to unpredictable external stimuli.	
Did the project design reference past lessons learned and incorporate them into decision-making processes?	JC1.9 The intervention design references design decisions made due to previous lessons learnt.	1 = Initial project designs did not consider any previous projects or lessons learnt. 2 = Initial project designs somewhat considered previous projects or lessons learnt. 3 = Initial project designs somewhat considered and implemented some previous projects or lessons learnt. 4 = Initial project designs fully considered previous projects and lessons learnt.	
Score for Relevance 2			
Coherence			
Internal Coherence			

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
To what extent do other AfDB interventions support or undermine the rail and aviation cluster projects and vice versa?	JC2.1 Other AfDB interventions and rail/aviation projects are mutually beneficial.	1 = There is no evidence of any consideration for mutual benefits between other AfDB interventions and rail/aviation projects. 2 = There are limited indications that some consideration was given to mutual benefits between other AfDB interventions and rail/aviation projects. 3 = Some effort was made to enable and deliver mutual benefits between other AfDB interventions and rail/aviation projects. 4 = There is clear evidence that significant consideration was given to ensuring mutual benefits between other AfDB interventions and rail/aviation projects, which resulted in some mutual benefits.	
To what extent were synergies between the AfDB's interventions in the transport sector and other AfDB interventions optimized (integrated solutions)?	JC2.2 The projects tried to establish synergies with other Bank's or other donors' interventions in the country/region.	1 = There is no evidence of AfDB interventions complementing, harmonizing, or coordinating with other AfDB programs and activities. 2 = Limited evidence exists of AfDB interventions complementing, harmonizing, or coordinating with other AfDB programs and activities. 3 = Some efforts were made by AfDB interventions to complement, harmonize, and coordinate with other AfDB programs and activities, adding value, but it was not fully realized. 4 = AfDB interventions are clearly demonstrated to complement, harmonize, and coordinate with other activities by the Bank, adding significant value to the overall development efforts.	
How did project management staff consider the opportunities and risks posed by AfDB or other donor projects initiated during project delivery?	JC2.3 The project management staff considered the opportunities and risks of AfDB or other donor projects which initiated during delivery.	1 = The Risk/Mitigation Measures in the Project Logical Framework section of the PAR were not considered during project implementation leading to complications in output delivery. 2 = Risk/Mitigation Measures were considered but not implemented effectively enough to avoid complications in output delivery. 3 = Risk/Mitigation Measures were considered and proactively implemented, avoiding complications in output delivery. 4 = Risk/Mitigation Measures were actively applied throughout the project, leading to smooth output delivery	
Score for Coherence 1			
Coherence			
External Coherence			

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
To what extent are the AfDB's interventions complementary, harmonized and coordinated with other development partners' support to RMCs, adding value while avoiding duplication of effort?	JC2.5 AfDB interventions complement, harmonize, and coordinate with other development partners, adding value.	1 = There is no evidence of AfDB interventions complementing, harmonizing, or coordinating with other development partners. 2 = Limited evidence exists of AfDB interventions complementing, harmonizing, or coordinating with other development partners. 3 = Some efforts were made by AfDB interventions to complement, harmonize, and coordinate with other development partners, adding value, but it was not fully realized. 4 = AfDB interventions are clearly demonstrated to complement, harmonize, and coordinate with other development partners, adding significant value to the overall development efforts.	
Score for Coherence 2			
Effectiveness			
Outputs Achievement			
To what extent were transport projects' outputs delivered as planned?	JC3.1 Project outputs were realised as expected.	1 = Less than 50% of the project outputs (as specified in the PAR and evaluated in the PCR and through analysis of other relevant documents and stakeholder interviews) have been delivered (or are on track to be delivered). 2 = Between 50% and 75% of the project outputs were delivered (or are on track to be delivered). 3 = Between 75% and 100% of the project outputs were delivered (or are on track to be delivered). 4 = All project outputs were delivered as planned.	
Score for Effectiveness 1			
Outcomes Achievement			
Can you provide evidence demonstrating that the project has effectively increased the movement of people and goods, enhanced multimodality, and improved access to markets?	JC3.2 Evidence shows that the project has increased movement of people and goods, improved multimodality and access to markets.	1 = There is no evidence that the project will ever achieve the intended connectivity objectives outcomes (as specified in the PAR and evaluated in the PCR and through analysis of other relevant documents and stakeholder interviews). 2 = Weak questionable evidence on improved connectivity. 3 = Limited but convincing evidence that connectivity has been improved. 4 = The project has achieved or exceeded all connectivity targets as specified in the PAR and PCR (traffic volume, access to markets).	
What evidence supports the claim that travel time and transport costs have been reduced as a result of the project's implementation?	JC3.3 Evidence suggests that travel time and transport costs have been reduced.	1 = No savings achieved. 2 = Savings significantly below expectations (less than 50%). 3 = Savings close to expectations (75%-100%). 4 = Expected savings fully achieved or beyond expectations.	

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
Where relevant (e.g. not for aviation): To what extent does the evidence indicate that the project has improved access to basic services such as healthcare, water, and education, while also creating job opportunities for all, including women and youth?	JC3.4 Evidence suggests that the project has improved access to basic services (healthcare, water, education) and contributed to creating job opportunities for all (including women and youth).	1 = There is no evidence that the project will ever achieve the intended basic services outcomes (as specified in the PAR and evaluated in the PCR and through analysis of other relevant documents and stakeholder interviews). 2 = Weak questionable evidence on improved living conditions of the local population. 3 = Limited but convincing evidence that the living conditions of the local populations has improved. 4 = The project has achieved or exceeded all targets.	
Based on available evidence, how has the project contributed to enhancing transport safety?	JC3.5 Evidence suggests that the project has contributed to improving transport safety.	1 = Rail/air traffic safety has worsened as a result of the project. 2 = Weak questionable evidence on improved rail/air traffic safety. 3 = Limited but convincing evidence that rail/air traffic is safer compared to the past. 4 = The project has achieved or exceeded all safety targets (as specified in the PAR and evaluated in the PCR).	
Is there evidence demonstrating that the project did not lead to unintended negative effects on local populations and the environment, and if such effects occurred, were they successfully mitigated?	JC3.6 Evidence suggests that the project has not generated unintended negative effects on the local populations (increased pollution, deforestation, HIV/AIDS transmission, security issues) and on climate and that when these effects occurred, they were successfully mitigated.	1 = The project had serious consequences on the local populations and on the climate, which were not foreseen in the ESMP. 2 = The projects had some unintended effects; when these occurred, the project could not address them satisfactorily. 3 = The projects had some minor unintended effects; when these occurred they were successfully addressed. 4 = The project had no unintended environmental, climate and social effects.	
How does the evidence indicate that the project had a positive impact on regional integration?	JC3.7 Evidence shows that the project had a positive impact on regional integration.	1 = The project did not bring any improvement in regional integration. 2 = The project only had minor impacts on regional integration. 3 = The project had some positive impacts on regional integration, but did not achieve the initial targets (as specified in PAR and measured in PCR). 4 = The project significantly improved regional integration (trade flows).	
Score for Effectiveness 2			
Efficiency			

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
Were delays present during the delivery of the project?	JC4.1 Evidence proves that projects did not suffer delays in the implementation.	1 = The ratio of planned implementation time (as per PAR) and actual project implementation time is expected to be <0.50. 2 = The ratio of planned implementation time (as per PAR) and actual implementation time is expected to be <0.75 and ≥0.50. 3 = The ratio of planned implementation time (as per PAR) and actual implementation time is expected to be <1 and ≥0.75. 4 = The ratio of planned implementation time (as per PAR) and actual implementation time is expected to be ≥1.	
Were there cost overruns, and were costs proportionate to the construction?	JC4.2 Evidence proves that there were no cost overruns and that resources were used efficiently.	1 = The ratio of the median percentage physical implementation of the project outputs and commitment rate (per the PCR) is <0.50. The project delivered significantly less outputs within the available budget. 2 = The ratio of the median percentage physical implementation of the project outputs and commitment rate is <0.75 and ≥0.50. The project delivered less outputs within the available budget. 3 = The ratio of the median percentage physical implementation of the project outputs and commitment rate is <1 and ≥0.75. The project largely delivered the outputs expected within the available budget. 4 = The ratio of the median percentage physical implementation of the project outputs and commitment rate is ≥1. The project delivered all or more outputs than expected within the available budget.	
How does the evidence demonstrate that the project was implemented as planned?	JC4.3 Evidence shows that the project was implemented as planned.	1 = The average rating of applicable IP criteria ratings (as per the PCR) is comprised between 1.0 and 1.49. Most dimensions of implementation processes have not been satisfactory which has jeopardized the achievement of project results. 2 = The average rating of applicable IP criteria ratings is comprised between 1.5 and 2.49. Several dimensions of implementation processes have not been satisfactory which has jeopardized the achievement of some project results. 3 = The average rating of applicable IP criteria ratings is comprised between 2.5 and 3.49. The implementation processes have for the most part been satisfactory and has for the most part lead to the anticipated. 4 = The average rating of applicable IP criteria ratings is comprised between 3.5 and 4. The implementation processes have for the most part been highly satisfactory and has to lead to the anticipated results.	

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
To what extent were the costs of the intervention funded by the AfDB commensurate to their planned benefits?	JC4.4 Evidence proves that project costs were commensurate to benefits.	1 = The ratio of the ERR at completion and the anticipated ERR at appraisal is <0.50. 2 = The ratio of the ERR at completion and the anticipated ERR at appraisal is <0.75 and ≥ 0.50. 3 = The ratio of the ERR at completion and the anticipated ERR at appraisal is <1 and ≥ 0.75. 4 = The ratio of the ERR at completion and the anticipated ERR at appraisal is ≥ 1.	
To what extent has AfDB's governance structure hindered project output delivery?	JC4.5 AfDB's governance processes and project output delivery.	1 = AfDB governance processes required extensive navigating, leading to major delays in project output delivery. 2 = AfDB governance processes required substantial navigating leading to substantial delays in project output delivery. 3 = AfDB governance processes required minimal navigating leading to minor delays in project output delivery. 4 = AfDB governance processes were efficient and well-designed, accelerating project output delivery.	
Score for Efficiency			
Sustainability			
Will the project continue to deliver at the same level in the future, and are systems and budgets in place to ensure this?	JC5.1 Evidence suggests that the project is financially sustainable.	1 = The project has not put in place any mechanisms for financial sustainability, and the flow of benefits associated with the project are not expected to continue after completion. 2 = The project marginally contributed to building or strengthening institutional capacities in road/port/cross-border rail and aviation management. Country systems and capacities remains weak to ensure the continued flow of benefits associated with the project after completion. 3 = The project significantly contributed to building or strengthening institutional capacities in road/port/cross-border rail and aviation management. Country systems and capacities are sufficient to ensure the continued flow of benefits associated with the project after completion. 4 = The project was critical in building or strengthening institutional capacities in road/port/cross-border rail and aviation management. Country systems and capacities are sufficient to ensure the continued flow of benefits associated with the project after completion.	

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
Is the project technically sound and does it build in any resilience to climate change?	JC5.2 Evidence suggests that the project is technically sound and resilient to climate change.	1 = There is a high likelihood that factors related to the technical design of the intervention may severely impact the achievement of the results. 2 = There is a substantial likelihood that factors related to the technical design of the intervention may adversely impact the achievement of the results. 3 = There is a relatively low likelihood that factors related to the technical design of the intervention may adversely impact the achievement of the results. 4 = There is a very low likelihood that factors related to the technical design of the intervention may adversely impact the achievement of the results.	
How compelling is the evidence suggesting that the project has contributed to enhancing the capacity to sustainably manage transport infrastructure?	JC5.3 Evidence suggests that the project has contributed to increasing the capacity to manage sustainably transport infrastructure.	1 = There is no evidence that the project will ever improve government capacity to deliver similar projects in the future. 2 = Weak questionable evidence on improved capacity to deliver similar projects in the future. 3 = Limited but convincing evidence that the project contributed to increase capacity to deliver similar projects in the future. 4 = Substantial improvements in transport sector governance and capacity to deliver similar projects in the future.	
Has the project fostered and enabled sustainable partnerships to forge and encouraged ownership of project results with beneficiaries?	JC5.4 Evidence suggests that the project has forged a sustainable partnership and ownership of project results with beneficiaries.	1 = The project has not been effective at involving the relevant stakeholders and there is no sense of ownership amongst the beneficiaries. No partnerships with relevant stakeholders have been established to ensure the continued maintenance and management of project outputs. 2 = The project has only involved a small number of stakeholders and there is limited ownership among beneficiaries. No partnerships with relevant stakeholders have been established to ensure the continued maintenance and management of project outputs. 3 = The project has been effective in involving most stakeholders and promoting a sense of ownership among beneficiaries. Partnerships with relevant stakeholders (local authorities, civil society organizations, private sector) have been established and are deemed sufficient enough to ensure the continued maintenance and management of project outputs. 4 = The project has been effective in involving all relevant stakeholders and there is a strong sense of ownership among beneficiaries. Partnerships with relevant stakeholders (local authorities, civil society organizations, private sector) have been established and stakeholders are dedicated deemed sufficient enough to ensure the continued maintenance and management of project outputs.	

Specific Questions	Judgement Criteria	Judgement Scores	Insert Score
To what extent has the AfDB assisted RMCs to appropriately assess and implement environmental/climate/social mitigation/enhancement measures?	JC5.5 Evidence shows that the project results are environmentally and socially sustainable.	1 = The Environmental and Social Management Plan (ESMP) has not been implemented; institutional capacity and funding are not available to ensure the environmental and social sustainability of the operation. 2 = The ESMP has been implemented with major delays or in an unsatisfactory manner; institutional capacity and funding are deemed insufficient to ensure the environmental and social sustainability of the operation. 3 = The ESMP has largely been implemented in a timely and satisfactory manner; institutional capacity and funding are deemed sufficient to ensure the environmental and social sustainability of the operation. 4 = The ESMP has largely been implemented in a timely and satisfactory manner; institutional capacity is strong and there is sufficient funding to ensure the environmental and social sustainability of the operation.	
Score for Sustainability			
Commentary			
What factors have affected progress towards the objectives and how are they linked to the AfDB intervention?	JC5.5 Factors affecting progress identified and addressed, optimizing project outcomes.		

Annex 3: References

- AfDB. 1993. AfDB's Transport Sector Policy.
- AfDB. 2008. AfDB Medium-Term Strategy 2008-2012.
- AfDB. 2012. AfDB Action Plan on Climate Change 2011-2015.
- AfDB. 2013. African Development Bank Group Ten-Year Strategy (TYS) 2013-22.
- AfDB. 2015. Rail Infrastructure in Africa (2015b).
- AfDB. 2015. Regional Integration Policy and Strategy 2014-23 - Integrating Africa: Creating the Next Global Market (2015c).
- AfDB. 2017. African Development Bank Climate Change Action Plan 2016-2020 (2017a).
- AfDB. 2017. Infrastructure and Urban Development Annual Report 2017 (2018b).
- AfDB. 2017. The Bank Group Results Measurement Framework 2016-2025 (2017c).
- AfDB. 2018. Infrastructure and Urban Development Annual Report 2018 (2019c).
- AfDB. 2019. African Development Bank Group's Policy on Non-Sovereign Operations.
- AfDB. 2019. Infrastructure and Urban Development Department - Annual Report 2019. African Development Bank - Building Today, a Better Africa Tomorrow. (2019b).
- AfDB. 2019. Cross-Border Road Corridors: The Quest to Integrate Africa (2019c).
- AfDB. 2021. The African Development Bank Group Gender Strategy 2021-2025.
- AfDB. 2022. Bank Policy for Addressing Fragility 2022-2026 (2022a).
- AfDB. 2022. 2nd Climate Change Action Plan (2016-2020) - Completion Report (2022b).
- AfDB. 2022. Infrastructure and Urban Development Annual Report 2020-21 (2022c).
- AfDB. 2022. AfDB Sustainable Urban Development Action Plan (SUDAP) (2022d)
- AfDB. 2023. Annual Development Effectiveness Review 2023 (2023a).
- AfDB. 2023. Cross-Border Road Corridors - Expanding Market Access in Africa and Nurturing Continental Integration (2023b).
- AfDB. 2023. Climate Change and Green Growth Strategic Framework: Operationalizing Africa's Voice - Action Plan 2021-2025. African Development Bank - Building Today, a Better Africa Tomorrow. (2023c).
- AfDB. 2023. Work Programme and Budget Document 2023-2025 - Strengthening the Bank to Support Africa's Economic Recovery and Resilience (2023d).
- African Union Commission. 2017. Transport Policy Paper Delivering Sustainable Transport for Realizing AU Agenda 2063.
- African Union Commission. 2020. Programme for Infrastructure Development in Africa (PIDA) 2021-30.
- African Union Commission 2021. The Third Ordinary Session of the Specialized Technical Committee on Transport, Transcontinental and Interregional Infrastructure, and Energy (STC-TTIIIE).
- African Union Development Agency (AUDA), & New Partnership for Africa's Development (NEPAD). 2022. Second Continental Report on the Implementation of Agenda 2063.
- African Union Development Agency (AUDA), & New Partnership for Africa's Development (NEPAD). 2023. PIDA: First 10-Year Implementation Report.
- Africa50. 2023. Annual Report.
- Asian Development Bank. 2023. Strategy 2030 Transport Sector Directional Guide. <https://www.adb.org/>
- Economic Commission for Africa. 2022. The African Statistical Yearbook 2021. African Development Bank - Building Today, a Better Africa Tomorrow.
- European Union, & African Union. 2021. Towards an enhanced Africa-EU Cooperation on Transport and Connectivity Report by the Task Force on Transport and Connectivity 2.
- Evaluation Cooperation Group (ECG). 2012. Good Practice Standards for the Evaluation of Public Sector Operations.
- IBRD/World Bank Group. 2020. Sub-Saharan Africa Transport Policy Program.
- IBRD/World Bank Group. 2023. Sub-Saharan Africa Transport Policy Program (SSATP).
- IDEV. 2014. Transport in Africa: The African Development Bank's Intervention and Results for the Last Decade Summary Evaluation Report.
- Islamic Development Bank. 2022. Annual Evaluation Report.
- Japan International Cooperation Agency. 2015. PEDEC-NACALA: Final Study Report.
- Organization for Economic Co-operation and Development (OECD) & African Centre for Economic Transformation. 2020. Quality Infrastructure in 21st Century Africa.
- Raitzer, D., Blöndal, N., & Sibal, J. (2019). Impact Evaluation of Transport Interventions: A Review of the Evidence.
- University of Arizona. 2000. What is Qualitative Comparative Analysis (QCA).

Endnotes

- 1 Highly Satisfactory (4), Satisfactory (3), Partly Unsatisfactory (2), and Unsatisfactory (1).
- 2 Nacala Rail Corridor and Port Project & Value Addition Project are two distinct, but complementary projects undertaken by the AfDB. Owing to their close relationship, these projects were evaluated together and are hence considered collectively in this report.
- 3 Highly Satisfactory (4), Satisfactory (3), Partly Unsatisfactory (2), and Unsatisfactory (1).
- 4 Such as the Support to AUC/AFCAC on Single African Air Transport Market.
- 5 Sharm El-Sheikh Airport Development Project and Jomo Kenyatta International Airport Airfield Expansion Project.
- 6 Feasibility Study on the Extension of the Cameroon-Chad Railway Project and Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway Project.
- 7 Of the 16 cluster projects, two were cancelled and one was terminated.
- 8 These include Corporate Loan to Ethiopian Airlines; Democratic Republic of Congo - Priority Air Safety Project Phase II; Feasibility Study on the Ethiopia-Sudan Standard Gauge Railway; Ghana Airports Company Limited (GACL) Capital Investment Programme; Morocco - Railway Infrastructure Reinforcement Project; South Africa - Corporate Loan to Transnet; and Tunisia - Rail infrastructure Modernization (II).
- 9 According to the Presidential Directive No. 02/2015, in force at the time of the evaluation period, the AfDB benchmarks implementation phases against specific reference points, thus two months for each of the first three phases (approval to signature; signature to effectiveness; and effectiveness to first disbursement) and six months from approval to first disbursement.
- 10 Project Assessment - Nacala Rail and Port Project & Malawi – Nacala Rail and Port Value Addition Project; Tunisian Railway Infrastructure Modernization (Phase II).
- 11 [Moody's, Standard & Poor's, Fitch and the Japanese Credit Rating Agency](#) have all assigned highly favourable credit ratings on the Bank's long-term senior- and subordinate debt due to positive outlooks on its financial condition, member support, capital adequacy and preferred creditor status.



IDEV

Independent Development Evaluation
African Development Bank



About this evaluation

Independent Development Evaluation (IDEV) conducted a cluster evaluation of 16 rail and aviation projects supported by the African Development Bank Group (“the Bank”) between 2012 and 2023 in its Regional Member Countries (RMCs). This evaluation complements the [cluster evaluation of road and port projects](#), and together, they provide specific inputs to IDEV’s broader evaluation of the [Bank’s support for the transport sector \(2012-2023\)](#) as a whole. The evaluation examined the projects’ relevance, coherence, efficiency, effectiveness, and sustainability, and the additionality of five non-sovereign projects.

The evaluation noted that the rail and aviation projects were aligned to the relevant strategies and priorities of the Bank and other relevant entities, and the needs of its RMCs. The cluster projects exhibited robust synergies and interlinkages with other projects in the indicative operational program of the Country Strategy Papers and/or the existing portfolio of the two sub-sectors, particularly through their support for regional integration. Although cluster projects displayed mixed output results, the evaluation found direct benefits for users in terms of connectivity and operational efficiency and, where relevant, in improvements to safety. The Bank’s involvement in the non-sovereign operations was found to have reduced political risk, enhancing buy-in from commercial clients. Institutional capacity-building projects were less successful in achieving their outcomes. Most projects were favorably assessed regarding their use of financial resources, but challenges related to project design, monitoring objectives, capacity limitations of executing agencies, inflexible Bank processes and requirements, and ineffective coordination were found to result in long delivery delays. Most infrastructure designs appeared to reflect appropriate quality standards and adaptation elements. However, long-term financial sustainability, implementation capacity, and environmental and social safeguard issues were noted.

The evaluation identified key lessons to enhance project performance, demonstrate the Bank’s additionality, ensure long-term financial sustainability, implement a robust monitoring and evaluation system, and enforce adequate environmental, social, and operational safeguards.



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